

Hybrid AC/DC Module Technical Notebook

Current C++ Architecture, Information Models, Mathematical Models, and Solver Workflows

HACDCPF C++ Module

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Contents

1 Scope and Documentation Contract

This notebook is the implementation-facing handbook for the current `hacdcpf` C++ module. It connects mathematical models to the public data structures, preprocessing steps, solver paths, optional build features, and result objects that are present in the repository now. It deliberately separates implemented runtime code from declared stubs and absent/planned APIs.

1.1 Implemented Runtime Families

The current notebook covers these implemented module families:

- the public model layer under `include/hacdcpf/model`;
- canonical projection and solver-data assembly in `src/model` and `src/power_flow`;
- hybrid AC/DC power flow through `solve_power_flow`;
- DC-only, FDPF, linearized AC/DC, adaptive/islanded, distributed-slack, HELM, and continuation PF entry points exposed in the power-flow headers;
- AC OPF through `solve_ac_opf`, including native reduced-space and parity full-space IPM routes;
- DC OPF through `solve_dc_opf`;
- reactive-power optimization through `solve_rpo`;
- carbon emissions analysis through `compute_carbon_analysis` (BFS proportional tracing + matrix solve, with optional PTDF weighting and sector-level attribution);
- time-series power flow pipeline through `run_time_series_power_flow` (multi-scenario, multi-horizon profiles with snapshot solve per time step);
- annual production simulation through `run_annual_production_simulation` (hourly PV/wind dispatch, curtailment accounting, energy yield statistics);
- lifecycle simulation through `run_lifecycle_simulation` (multi-year degradation, replacement scheduling, levelised cost computation);

- JSON, MATPOWER/JPC, case-builder, PF result JSON, OPF result JSON, and carbon result JSON support;
- optional Excel workbook I/O when `HACDCPF_ENABLE_EXCEL=ON`;
- optional OpenDSS bridge source inclusion when `HACDCPF_ENABLE_OPENDSS=ON`.

1.2 Current Stub and Absent Boundaries

Area	Current code status
<code>analysis::solve_distribution_power_flow</code>	Declared and compiled, but <code>src/power_flow/</code> returns a non-converged result with <code>status="not_implemented"</code> .
Three-stage MILP reliability	Public API and data types defined in <code>include/hacdcpf/analysis/three_stage_reliability.hpp</code> ; <code>src/reliability/three_stage_reliability.cpp</code> is a placeholder that returns a graceful <code>notyet_implemented</code> error. The native C++ MILP solver is under development.

1.3 Documentation Principle

This notebook follows four rules.

1. Equations are written in the same sign convention as the current C++ implementation.
2. Field explanations are tied to concrete structs, not generic textbook entities.
3. Each component is classified by how it enters a module: direct state, fixed injection, projected equivalent, or not yet consumed.
4. Extension points declared in headers are explicitly separated from functionality that is already wired in source files.

1.4 Canonical Modeling Pipeline

The steady-state solvers do not consume every rich component table directly. Instead, the dominant runtime path is:

1. start from a public `HybridPowerSystem`,
2. call `project_to_canonical_models`,
3. build `SolverData` through `make_solver_data`,
4. assemble residuals, Jacobians, or optimization models for PF and OPF.

The projection stage currently performs the following transformations:

- `FlexibleLoad` $\rightarrow$ equivalent `Load`,
- `AsymmetricLoad` $\rightarrow$ equivalent aggregated `Load`,
- `Transformer2W` and `Transformer3W` $\rightarrow$ one or more equivalent `ACBranch` objects,
- `Switch` $\rightarrow$ equivalent `ACBranch` whose `in_service` flag follows the switch state,

- `CircuitBreaker` → equivalent `ACBranch`, followed by zero-impedance bus merging when switches or circuit breakers are present,
- `Charger` → accumulation into `ChargingStation` demand,
- `VirtualPowerPlant` and `Microgrid` → equivalent `AC StaticGenerator` rows,
- `MobileStorage` → equivalent `AC Storage` at its current bus,
- `EnergyRouter` → internal DC buses plus VSC/DCDC equivalents,
- optional `ThreePhaseACSystem` → canonical single-phase-equivalent AC models when the explicit AC system is absent.

1.5 Important Boundary Between “Implemented” and “Declared”

The parity OPF formulation now instantiates **all** enhanced component decision blocks as run-time variables with full Jacobian and Hessian wiring:

- generator dispatch (P_g, Q_g) ,
- VSC converter power $(P^{\text{conv,ac}}, Q^{\text{conv,ac}}, P^{\text{conv,dc}})$,
- DCDC converter input power P^{dcdc} with analytical I^2R loss derivatives,
- DC bus voltages v^{dc} ,
- per-bus load shedding $(\Delta P_d, \Delta Q_d)$ with VOLL penalty,
- curtailable renewable dispatch $(P^{\text{ren}}, Q^{\text{ren}})$ for `RenewableGen` with `curtailable=true` and `PVSystem` with `controllable=true`,
- storage dispatch $(P^{\text{stor}}, Q^{\text{stor}})$ for AC storage and $P^{\text{stor,dc}}$ for DC storage,
- flexible load adjustment P^{flex} .

These are no longer extension hooks: they are wired decision variable blocks with explicit mapping arrays in `parity_formulation.cpp`. The OPF JSON serializer preserves the enhanced dispatch arrays as both legacy bare arrays and structured dispatch objects with identity maps.

2 Nomenclature

2.1 Notation Policy

This notebook uses a strict notation policy consistent with the C++ module:

- scalar variables use italic math symbols, for example P_i, Q_i, V_i ,
- sets use calligraphic symbols, for example $\mathcal{N}_{ac}, \mathcal{E}_{dc}$,
- all semantic qualifiers use upright superscripts via $\cdot$, for example $P_i^{\text{spec}}, P_i^{\text{calc}}, V_k^{\text{dc}}$,
- for variable symbols, subscripts are reserved for ordered indices (bus/component/time) and not for semantic class labels,
- set symbols may use compact domain tags in subscript, e.g., $\mathcal{N}_{ac}, \mathcal{E}_{dc}$,
- model-family and semantic words must stay in superscripts: write $P^{\text{dcdc}}, P^{\text{flex}}, P_r^{\text{rated}}, c_r^{\text{curt}}$, and $\mathcal{R}^{\text{curt}}$.

2.2 Index and Qualifier Conventions

Token	Meaning
i, j	AC bus indices
k, m	DC bus indices
c	converter index (VSC / DCDC)
b	branch index
g	generator index
spec	specified (assembled) quantity from component injections/demands
calc	network-calculated quantity from current state
ac, dc	AC-side / DC-side quantity qualifier
conv	converter-side quantity qualifier
dcdc, flex	DCDC-dispatch / flexible-load variable class qualifier
rated, curt	rated-capacity / curtailment-related qualifier
min, max	lower/upper operating bounds

2.3 Canonical Naming Grammar

Use the canonical symbol template

$$X_{\text{indices}}^{\text{qualifiers}}, \quad (1)$$

with the following rules:

- index block ($\cdot$): ordered indices only ($i, j, k, c, r, t, \dots$),
- qualifier block ($\cdot$): semantic tags only (spec, calc, conv, dcdc, flex, rated, curt, min, max, ...),
- when multiple qualifiers are needed, order them as: model/class $\rightarrow$ side or direction $\rightarrow$ status/bound/cost.

Examples:

$$P_i^{\text{conv,ac}}, P_k^{\text{dcdc,in}}, P_r^{\text{rated}}, c_r^{\text{curt}}. \quad (2)$$

2.4 Superscript/Subscript Rule

Rule	Usage
Subscript ($\cdot$)	for variable symbols: ordered indices only (i, j, k, c, r, t); set symbols may use domain tags ($\mathcal{N}_{ac}, \mathcal{E}_{dc}$).
Superscript ($\cdot$)	semantic labels: domain, role, operating mode, variable class, bounds, and status qualifiers.
Forbidden mixed style	avoid semantic words in subscript, such as $P_{dcdc}, P_{flex}, \mathcal{R}_{curt}, P_{rated}, c_{curt}$.
Preferred style	$P^{\text{dcdc}}, P^{\text{flex}}, \mathcal{R}^{\text{curt}}, P_r^{\text{rated}}, c_r^{\text{curt}}$.

2.5 Integration Status Codes

The integration matrix uses these codes:

- D: direct model consumption in the module,
- F: fixed-only consumption (no explicit variable),
- P: consumed after projection to canonical equivalents,
- N: not consumed by the module.

3 Current C++ Architecture

This section describes the repository as it exists in the current C++ module. The library target is `hacdcpf`, configured from the top-level `CMakeLists.txt` as a C++20 static library. The build links against `mipsolvers::mipsolvers`; that dependency supplies the shared engine types, solver adapters, and numerical backends used by the OPF and RPO code.

3.1 Source Layout

Directory	Current role
<code>include/hacdcpf/model</code>	Public data structures for AC, DC, converter, switching, three-phase, and result models.
<code>include/hacdcpf/api</code>	High-level public API facade in <code>hacdcpf.hpp</code> .
<code>include/hacdcpf/power_flow</code> and <code>src/power_flow</code>	Hybrid Newton PF, DC PF, FDPF, adaptive/islanded/distributed-slack variants, HELM, continuation, Jacobian assembly, PF utilities, <i>and</i> distribution network (BFS/3-phase) power flow.
<code>include/hacdcpf/optimal_power_flow</code> and <code>src/optimal_power_flow</code>	AC OPF routing, parity formulation, parity IPM, DC OPF, and reactive-power optimization.
<code>include/hacdcpf/power_models</code> and <code>src/power_models</code>	Builder-style models for AC OPF, AC/DC OPF, DC OPF, LinDistFlow, and SCUC experiments.
<code>include/hacdcpf/io</code> and <code>src/io</code>	MATPOWER/JSON/JPC case I/O, optional Excel workbook I/O, optional OpenDSS bridge, and case builders.
<code>include/hacdcpf/carbon_analysis</code> and <code>src/carbon_analysis</code>	Carbon emissions tracing (<code>run_carbon_analysis</code>).

Directory	Current role
<code>include/hacdcpf/time_series</code> and <code>src/time_series</code>	UC/OPF/PF multi-period time-series pipeline (<code>solve_time_series_pf</code> , <code>solve_unit_commitment</code>); annual production simulation via hierarchical temporal decomposition (<code>solve_annual_production_simulation</code>); multi-year lifecycle simulation with degradation and NPV (<code>run_lifecycle_simulation</code> , <code>run_lifecycle_comparison</code>); shared <code>UCSchedule</code> , <code>TimeSeriesPFResult</code> , and <code>TimeSeriesData</code> types.
<code>include/hacdcpf/reliability</code> and <code>src/reliability</code>	Monte Carlo reliability assessment (LOLP, EENS, SAIDI/SAIFI).
<code>include/hacdcpf/resilience</code> and <code>src/resilience</code>	Distribution resilience assessment with microgrid islanding and storage dispatch (<code>run_distribution_resilience_assessment</code>).
<code>include/hacdcpf/engine</code>	Compatibility/forwarding headers into <code>mipsolvers::engine</code> , plus the <code>hacdcpf</code> Newton PF class declaration and the <code>detail/core_compat.hpp</code> backward-compat aliases. There is no <code>src/engine</code> directory in this repository.

3.2 Top-Level Public API

The facade header `include/hacdcpf/api/hacdcpf.hpp` currently exposes:

- PF: `solve_power_flow`, `solve_dc_power_flow`, `solve_power_flow_adaptive`, `solve_power_flow_islanded`, `solve_power_flow_distributed_slack`, `solve_power_flow_distributed_slack_full`, `solve_power_flow_fdpf`, and `solve_ac_dc_power_flow`;
- persistent PF handles: `create_solver_handle`, `reset_solver_handle`, `solve_handle`, `solve_dc_handle`, and `destroy_solver_handle`;
- OPF: `solve_ac_opf`, `solve_dc_opf`, and `solve_rpo`;
- time-series pipeline: `solve_time_series_pf` and `solve_unit_commitment`;
- carbon analysis: `run_carbon_analysis`;
- annual production simulation: `solve_annual_production_simulation`;
- lifecycle simulation: `run_lifecycle_simulation`, `run_lifecycle_comparison`, and `apply_capacity_scaling`;
- reliability: `run_reliability_assessment`;
- resilience: `run_distribution_resilience_assessment`;
- distribution PF: `solve_distribution_power_flow` and `solve_three_phase_nr`.

Lower-level namespaces expose additional solver objects and helper entry points. The power-flow namespace includes `solve_hybrid`, `solve_hybrid_adaptive`, `solve_ac`, `solve_dc`, `FDPFSolver`, `AdaptiveSolver`, `DistributedSlackSolver`, `HelmSolver`, `CpfSolver`, `HomotopyContinuationSolver`, and the header-only `solve_three_phase` wrapper.

3.3 Build-Time Optional Features

CMake option	Default	Current effect
HACDCPF_BUILD_TESTS	ON	Enables the Catch2 test targets under <code>tests/</code> .
HACDCPF_ENABLE_EXCEL	OFF	Adds <code>src/io/excel_io.cpp</code> ; otherwise <code>HACDCPF_NO_EXCEL</code> is defined and the public Excel functions throw.
HACDCPF_ENABLE_OPENDSS	OFF	Adds <code>src/io/openss_bridge.cpp</code> ; otherwise OpenDSS bridge functionality is absent from the build.

3.4 Algorithm Index

The notebook uses the following code-aligned algorithm labels.

Label	Main implementation	Purpose
A1	<code>src/model/network_utils.cpp</code>	Rich-model projection to canonical solver inputs.
A2	<code>src/power_flow/solver_data.cpp</code>	<code>SolverData</code> construction, injection aggregation, and matrix build.
A3	<code>src/power_flow/newton_solver.cpp</code>	Hybrid AC/DC Newton power flow.
A4	<code>src/power_flow/fdpf_solver.cpp</code>	AC fast-decoupled power flow.
A5	<code>src/power_flow/dc_solver.cpp</code>	DC-only power flow.
A6	<code>src/power_flow/adaptive_solver.cpp</code>	Island-aware PF orchestration.
A7	<code>src/power_flow/distributed_slack_solver.cpp</code>	Simplified and full distributed-slack PF.
A8	<code>src/optimal_power_flow/ac_opf.cpp</code>	AC OPF top-level routing and fallback selection.
A9	<code>src/optimal_power_flow/parity_formulation.cpp</code>	Full-space parity NLP construction and derivatives.
A10	<code>src/optimal_power_flow/parity_ipm.cpp</code>	Mehrotra/Gondzio-style parity IPM solve loop.
A11	<code>src/optimal_power_flow/dc_opf.cpp</code>	DC OPF QP/LP model build and backend dispatch.
A12	<code>src/optimal_power_flow/reactive_power_opt.cpp</code>	Reactive-power optimization by OA/coordinate search.
A13	<code>src/power_flow/homotopy_continuation.cpp</code>	Continuation Power Flow (CPF) predictor–corrector (§??).
A14	<code>src/power_flow/homotopy_continuation.cpp</code>	Homotopy continuation fallback solver (§??).
A15	<code>src/power_flow/newton_krylov.cpp</code>	Newton–GMRES Krylov fallback with Schur preconditioner (§??).

Label	Main implementation	Purpose
A16	<code>src/power_models/lindistflow_builder.cpp</code>	Linearised DistFlow LP builder (§??).
A17	<code>src/power_models/acopf_builder.cpp</code>	AML AC OPF model builder (§??).
A18	<code>src/power_models/acdcopf_builder.cpp</code>	AML hybrid AC/DC OPF model builder (§??).
A19	<code>src/power_models/scuc_builder.cpp</code>	MILP Security-Constrained Unit Commitment builder (§??).
A20	<code>src/time_series/annual_production_sim.cpp</code>	Hierarchical temporal decomposition annual production simulation: L0 annual plan → L1 monthly → L2 weekly UC → L3 daily OPF/PF replay.
A21	<code>src/time_series/lifecycle_simulation.cpp</code>	Multi-year lifecycle simulation with storage SoH degradation, renewable derating, NPV cost roll-up, and Tier-1/2/3 theoretical carbon error bounds.
A22	<code>src/network_reconfiguration/topology_reconfiguration.cpp</code>	Optimal Network Reconfiguration (ONR): LinDistFlow MILP build (A16), HiGHS B&C solve, spanning-tree validation, Newton PF verification (§??).
A23	<code>src/analysis/short_circuit.cpp</code>	Short-circuit Z-bus analysis: assemble $\mathbf{Y}_{\text{fault}}$, SparseLU factorize, per-bus Thévenin solve, IEC 60909 fault-current and fault-level computation (§??).

4 Engine Solver System

The `hacdcpf::engine` namespace in this repository is mainly a compatibility facade over `mipsolvers::engine`. The local headers keep historical include paths stable while importing engine symbols with `using` declarations. Current implementation files for LCQP, dual simplex, branch and cut, and external adapters live in the `mipsolvers` dependency, not under `src/engine` in this repository.

4.1 Forwarded Engine Surface

Local forwarding header	Imported symbols / role
<code>engine/problem_types.h</code> <code>pp</code>	Problem-class enums, symbolic model structs, LP/QP/MIP/NLP/MINLP model types, <code>SolveStats</code> , and <code>SolveResult</code> .
<code>engine/kernel/ipm/lcqp_solver.hpp</code>	<code>LCQPOptions</code> and <code>NativeLCQPAdapter</code> .

Local forwarding header	Imported symbols / role
<code>engine/kernel/lp_kernel/dual_simplex.hpp</code>	Simplex options, basis/result types, <code>StandardForm</code> LP, and <code>solve_lp_with_basis</code> .
<code>engine/kernel/linear_algebra/linear_solver.hpp</code>	Sparse solver abstraction and Eigen-based sparse solver factories.
<code>engine/kernel/globalization/globalization.hpp</code>	Dogleg, trust-region update, and pseudo-transient continuation helpers.
<code>engine/solver/external/adapters.hpp</code>	<code>HighsAdapter</code> , <code>IpoptAdapter</code> , <code>ScipAdapter</code> , and <code>GurobiAdapter</code> .
<code>engine/branch_and_cut.hpp</code>	Branching and node-selection enums plus B&C statistics/result/option types.

There are no local PDLP or LP-IPM backends documented by the current headers. The backends visible through this compatibility layer are LCQP, dual simplex, HiGHS, Ipopt, SCIP, and Gurobi, subject to the configuration of `mipsolvers` and the optional external packages available at build time.

4.2 Local Newton Solver Class

The one local engine-style solver declaration is `include/hacdcpf/engine/solver/native/nle/newton_solver.hpp`. Its implementation is `src/power_flow/newton_solver.cpp`; it is the hybrid PF Newton class used by the power-flow module. The file is not an engine implementation directory and does not define the optimization-engine LCQP or LP backends.

4.3 OPF Use of Engine Facilities

Consumer	Engine dependency	Current behavior
DC OPF	LCQP, Gurobi, HiGHS, dual simplex	<code>Auto</code> tries <code>NativeQP</code> , then Gurobi, then HiGHS, then native dual simplex. LMP arrays are currently filled with zeros because dual extraction is marked TODO in <code>dc_opf.cpp</code> .
RPO	<code>BCStats</code> only for result reporting	The current source runs an OA/coordinate-search workflow and reuses B&C statistics fields for counters; it does not call a generic branch-and-bound solve loop.
Parity AC OPF	local parity IPM	Uses the local KKT assembly and Eigen dense/sparse factorizations in <code>parity_ipm.cpp</code> .
Native AC OPF	local reduced-space KKT loop	Uses custom KKT assembly and factorization in <code>ac_opf.cpp</code> .

5 Native NLP and Interior-Point Details

The nonlinear OPF solvers are local to `src/optimal_power_flow`. They do not dispatch through an external NLP engine by default. The AC OPF entry point selects either a native

reduced-space primal-dual path or the parity full-space IPM path described in Section ??.

5.1 Native AC OPF Loop

The native path in `ac_opf.cpp` uses variables for non-slack AC angles, PQ voltage magnitudes, generator dispatch, DC voltage magnitudes, and VSC converter powers. Equality rows cover AC balances, DC balances, and VSC energy balance. Bound constraints are handled with barrier terms, while nonlinear branch and converter limits enter through exterior penalty terms. If this path fails and `ACOPFOptions::allow_fallback` is true, the code falls back to the fast dispatch plus AC PF projection route.

5.2 Parity IPM Loop

The parity path builds explicit equality and inequality rows in `parity_formulation.cpp` and solves the resulting full-space NLP in `parity_ipm.cpp`. The solve loop uses a Mehrotra predictor-corrector step with second-order complementarity correction, iterative refinement, and fraction-to-boundary primal/dual step lengths. Current centering is

$$\sigma = \left(\frac{\mu_{aff}}{\mu_{cur} + 10^{-16}} \right)^3 \quad (3)$$

after clamping the ratio to $[0, 1]$. The implementation also applies up to two extra Gondzio corrector solves when enabled by the current loop logic.

The KKT system is solved with Eigen `SparseLU` on sparse systems or `PartialPivLU` on dense systems. There is no separate local NLP/IPM engine source tree beneath `src/engine`.

6 Native IPM Versus Parity IPM

The two AC OPF nonlinear routes differ mainly in variable coverage and in how constraints are represented.

Feature	Native reduced-space path	Parity full-space path
Primary files	<code>ac_opf.cpp</code>	<code>parity_formulation.cpp</code> , <code>parity_ipm.cpp</code>
AC bus variables	non-slack angles and PQ magnitudes	full AC angle and magnitude blocks with bounds
Generator dispatch	P_g, Q_g variables	P_g, Q_g variables
VSC converters	explicit P^{ac}, Q^{ac}, P^{dc} variables and balance rows	explicit P^{ac}, Q^{ac}, P^{dc} variables and balance rows
DC bus voltages	variables, with tight bounds for <code>DC_V</code> buses	variables with DC balance rows and DC branch limits
Load shedding	not modeled	$\Delta P_d, \Delta Q_d$ variables with VOLL penalty
Curtable renewable / PV	fixed injections	packed decision vectors <code>pren_mw</code> , <code>qren_mvar</code> with identity maps
Storage	fixed injection	AC and DC storage active dispatch packed into <code>pstor_mw</code> ; AC storage also has <code>qstor_mvar</code>

Feature	Native reduced-space path	Parity full-space path
DCDC converters	not modeled as decision variables	input-side <code>pdc_dc_mw</code> variables folded into DC bus balance physics
Flexible loads	projected/fixed demand	<code>pflex_mw</code> variables with identity maps
Failure fallback	fast dispatch plus AC PF if allowed	native path if allowed, then native fallback behavior

The parity result stores enhanced component outputs in packed arrays. The maps `ren_map`, `stor_map`, `dc_dc_map`, and `flex_map` must be used to join those arrays back to the original component collections. For storage, `source_type=0` denotes AC storage and `source_type=1` denotes DC storage.

One implementation quirk is important for code review: `ConstraintIndex` still allocates `n_dc_dc_bal` and `i_dc_dc_bal`, and `build_problem` includes those rows in `n_eq_total`. The current equality evaluator leaves the dedicated DCDC rows unpopulated; DCDC physics is represented instead through the DC bus balance rows.

7 Symbols, Indexing, and Sign Convention

7.1 Electrical Bases

$$S_{base} = \text{system apparent-power base (MVA)}, \quad (4)$$

$$V_{base} = \text{nominal line-line voltage base (kV)}, \quad (5)$$

$$Z_{base} = \frac{V_{base}^2}{S_{base}}, \quad (6)$$

$$I_{base} = \frac{S_{base}}{\sqrt{3}V_{base}}. \quad (7)$$

7.2 Sets and Indexing

$$\mathcal{N}_{ac} : \text{AC buses}, \quad \mathcal{E}_{ac} : \text{AC branches}, \quad (8)$$

$$\mathcal{N}_{dc} : \text{DC buses}, \quad \mathcal{E}_{dc} : \text{DC branches}, \quad (9)$$

$$\mathcal{N}_{sl} : \text{AC slack buses}, \quad \mathcal{N}_{PQ} : \text{AC PQ buses}, \quad (10)$$

$$\mathcal{N}_{PV} : \text{AC PV buses}, \quad \mathcal{N}_{sl}^{dc} : \text{DC voltage-controlled buses}. \quad (11)$$

The public model stores external bus identifiers in each component (`bus`, `from_bus`, `to_bus`). Most solver kernels then remap those identifiers into contiguous zero-based local indices.

7.3 Core AC and DC State Variables

$$V_i = v_i e^{j\theta_i}, \quad (12)$$

$$Y_{ij} = G_{ij} + jB_{ij}, \quad (13)$$

$$V_k^{dc} = v_k^{dc}, \quad (14)$$

$$P_k^{dc,calc} = v_k^{dc} (G_{dc} v_k^{dc})_k. \quad (15)$$

For the main Newton power-flow path, the state vector is

$$x_{PF} = \left[\theta_{\mathcal{N}_{ac} \setminus \mathcal{N}_{sl}} \quad v_{\mathcal{N}_{PQ}} \quad v_{\mathcal{N}_{dc} \setminus \mathcal{N}_{sl}}^{dc} \right]^\top. \quad (16)$$

For the native primal-dual AC OPF path, the decision vector is extended to include generator dispatch and converter operating points:

$$x_{OPF} = \left[\theta \quad v \quad P_g \quad Q_g \quad v^{dc} \quad P^{\text{conv},ac} \quad Q^{\text{conv},ac} \quad P^{\text{conv},dc} \right]^\top, \quad (17)$$

with additional load-shedding variables in the parity formulation.

7.4 Unified Sign Convention: Bus Injection Positive

The current C++ module uses one consistent notation in the PF kernels and the notebook adopts the same rule:

Positive active or reactive power at a bus means injection into the network from the component attached to that bus.

Loads therefore store positive demand values in their data model, but they enter power-balance equations with a minus sign.

Quantity / field	Positive value means	Consequence in equations
<code>Generator::pg_mw, qg_mvar</code>	generator injects active/reactive power into its AC bus	added to $P_i^{\text{spec}}, Q_i^{\text{spec}}$.
<code>Load::p_mw, q_mvar</code>	consumption requested from the network	subtracted from $P_i^{\text{spec}}, Q_i^{\text{spec}}$; ZIP dependence scales the subtraction.
<code>Storage::p_mw</code>	discharge to the grid; charging is negative P	treated as generation when positive, load when negative.
<code>Microgrid::p_exchange_mw</code>	export from the microgrid to the host AC grid	added as AC-bus injection in PF; ignored when the microgrid is islanded.
<code>VSCConverter</code> AC-side $P^{\text{conv},ac}$	converter injects active power into the AC bus	positive AC injection must be balanced by negative DC-bus injection plus losses.
<code>VSCConverter</code> DC-side $P^{\text{conv},dc}$	converter injects active power into the DC bus	the current code enforces $P^{\text{conv},ac} + P^{\text{conv},dc} + P^{\text{loss}} = 0$.

Quantity / field	Positive value means	Consequence in equations
<code>DCDCConverter::p_ref_mw</code>	positive transfer from <code>bus_in</code> toward <code>bus_out</code>	implemented as negative injection at <code>bus_in</code> and positive injection at <code>bus_out</code> .
<code>BranchFlow::pf_mw</code>	active power sent from <code>from_bus</code> to <code>to_bus</code>	reporting only; no sign reversal in stored result.

7.5 Power-Balance Template

With bus-injection-positive notation, the balance equations are written as

$$\Delta P_i = P_i^{\text{spec}} - P_i^{\text{calc}}, \quad (18)$$

$$\Delta Q_i = Q_i^{\text{spec}} - Q_i^{\text{calc}}, \quad (19)$$

$$\Delta P_k^{\text{dc}} = P_k^{\text{dc,spec}} - P_k^{\text{dc,calc}}. \quad (20)$$

The specified term is assembled by adding all component injections and then subtracting all demand terms. This is the convention followed by the PF residual evaluator, the native AC OPF assembly, and the parity formulation.

8 Information Models and Field Dictionaries

8.1 Modeling Layers Used by the Code

The code base uses three distinct model layers.

1. **Public information model:** the rich structs under `include/hacdcpf/model/` used by I/O, case builders, and external callers.
2. **Canonical steady-state model:** the projected subset generated by `project_to_canonical_models`, where rich components are converted into equivalent loads, branches, or aggregated injections.
3. **Solver assembly model:** `powerflow::SolverData`, which stores contiguous vectors, aggregated per-bus injections, ZIP coefficients, and sparse matrices needed by the PF and OPF kernels.

8.2 System Containers and Solver-Level Metadata

System containers

Struct / group	Fields and meaning	Primary consumers
ACSystem	buses, branches, generators, static_generators, loads, flexible_loads, asymmetric_loads, shunts, storage, renewable_gens, pv_systems, external_grids, transformers_2w, transformers_3w, switches, circuit_breakers, charging_stations, chargers, motors; plus base_mva, freq_hz, name.	PF, OPF, I/O, projection
DCSystem	buses, branches, loads, storage, static_generators, dc_static_generators, pv_arrays, dc_circuit_breakers, plus base_mva and name.	PF, OPF, I/O
HybridPowerSystem	Holds ac, dc, vsc_converters, dcdc_converters, energy_routers, mobile_storage, vpps, microgrids, global base_mva, name, optional three_phase_ac, and optional provenance maps bus_merge_map and branch_expand_map.	all top-level APIs
ThreePhaseACSystem	Optional unbalanced AC subsystem with three-phase buses, lines, transformers, loads, generators, and external grids. It is projected to canonical AC objects when the explicit ACSystem is empty.	projection, PF, I/O

Power-flow options and results

Struct / group	Fields and meaning	Primary consumers
InitialState	vm, va, vdc provide optional warm starts for AC magnitude, AC angle, and DC voltage states.	Newton PF, DC PF
ReactiveLimit	qmin, qmax define reactive limits used by adaptive/PV-PQ logic.	adaptive PF extensions
DistributedSlack	participating_buses, participation_factors, reference_bus, and optional max_participation_p describe how slack mismatch is redistributed.	distributed-slack PF
PowerFlowOptions	Iteration and tolerance controls: max_iter, tol; robustness controls: line search, regularization, and step clipping; PV/PQ and converter hysteresis parameters; ZIP weights; solver profiling flag; loss_model; FDPF iteration cap.	PF family
SolverProfiling	Sparse-factorization counters, line-search and regularization statistics, PV/PQ switches, converter-mode switches, timing vectors, and cumulative timing totals.	PF result diagnostics
PowerFlowResult	Solved states vm, va, vdc; status fields converged, iterations, residual; and derived reports branch_flows, vsc_transfers, dcdc_transfers, er_port_transfers.	PF family outputs

Struct / group	Fields and meaning	Primary consumers
<code>AdaptiveSolverResult</code> , <code>IslandedSolverResult</code> , <code>DistributedSlackResult</code>	Variants of PF results that additionally expose detected <code>islands</code> , distributed-slack allocations, and buses hitting slack-sharing limits.	adaptive / islanded / distributed-slack PF

Analysis and OPF options/results

Struct / group	Fields and meaning	Primary consumers
<code>ACOPFOptions</code>	Inner/outer iteration caps, feasibility/stationarity tolerances, barrier schedule, merit-penalty and regularization terms, line-search settings, parallel AC evaluation threads, and path-selection flags <code>enable_primal_dual</code> , <code>use_parity_ipm</code> , <code>allow_fallback</code> .	AC OPF
<code>ACOPFResult</code>	Voltage solution, generator dispatch, optional bus-level load shedding, converter operating points, convergence metrics, status text, and profiling summary.	AC OPF output
<code>DCOPFOptions</code> , <code>DCOPFResult</code>	DC approximation controls, backend selector, branch-limit flag, load-shedding flag, voltage angles, generator dispatch, branch flows, load shedding, zero-filled LMP/vector dual placeholders, and solver status.	DC OPF
<code>RPOOptions</code> , <code>RPOResult</code>	Reactive-power objective settings, search limits, voltage/generator before/after vectors, tap and shunt changes, loss metrics, status, and B&C-style statistics fields reused as counters.	RPO
Analysis result types	Resilience, reliability (MC), short-circuit, ONR, time-series, and carbon are fully implemented. Distribution power flow (<code>solve_distribution_power_flow</code>) returns a stub result. The three-stage MILP reliability solver is under development; see Section ?? for the current stub boundary.	analysis declarations

`powerflow::SolverData`

Group	Fields and meaning	Primary consumers
Canonical component tables	Copies of projected AC/DC buses, branches, generators, converters, DCDC converters, energy routers, loads, DERs, shunts, switches, charging stations, external grids, VPPs, microgrids, and mobile storage.	PF, native OPF
Aggregated steady-state injections	<code>pg</code> , <code>qg</code> hold per-bus generation after aggregation of generators, static generators, renewables, PV, storage, VPPs, microgrids, and mobile storage.	PF, native OPF
Demand model data	<code>has_component_loads</code> , <code>pd_pu</code> , <code>qd_pu</code> , and per-bus ZIP coefficient vectors describe the demand model after load-table aggregation.	PF, native OPF
Network matrices	<code>ybus</code> and <code>gdc</code> store the canonical AC admittance and DC conductance matrices; <code>build_id</code> is used to validate Jacobian-pattern caching.	PF, native OPF
Bookkeeping	<code>base_mva</code> , <code>loss_model</code> , and global ZIP weights copied from <code>PowerFlowOptions</code> .	PF, native OPF

8.3 Component Inventory

Table ?? enumerates the rich public models currently present in the repository.

Table 15: Public component-model inventory

Struct	Mathematical role	Current steady-state integration mode
<code>ACBus</code>	AC state host, voltage limits, embedded bus demand and shunt	direct
<code>ACBranch</code>	AC series/shunt admittance, tap and phase-shift branch	direct
<code>Transformer2W</code>	named two-winding transformer with short-circuit/tap data	projected to <code>ACBranch</code>
<code>Transformer3W</code>	explicit three-winding transformer with pairwise short-circuit data	projected to three <code>ACBranch</code> equivalents
<code>ExternalGrid</code>	slack source and source-equivalent metadata	direct PF source metadata; richer fault fields are stored but no public SC solver is present
<code>AsynchronousMotor</code>	motor fault-contribution metadata	stored by the model and I/O; no public SC solver currently consumes it
<code>Generator</code>	synchronous generator dispatch and limits	direct

Struct	Mathematical role	Current steady-state integration mode
StaticGenerator	fixed/scaled distributed generation	direct fixed injection
RenewableGen	renewable injection with curtailment metadata	fixed injection in PF/native OPF; parity OPF decision variable when curtailable
PVSystem	PV inverter and array metadata	fixed injection in PF/native OPF; parity OPF decision variable when controllable
Load	AC demand with ZIP and motor-fraction metadata	direct
FlexibleLoad	demand-response model	projected to Load for PF; parity OPF has packed flexible-load variables
AsymmetricLoad	per-phase aggregate demand	projected to Load
Shunt	fixed or stepped shunt element	direct fixed shunt in PF; switchable steps are RPO discrete controls
Switch	switching device with electrical/protection data	projected to ACBranch for PF/OPF
CircuitBreaker	high-voltage breaker on an AC bus pair	projected to ACBranch ; can trigger zero-impedance bus merging
ChargingStation	aggregate EV charging demand	direct fixed load
Charger	individual EV charger	aggregated into ChargingStation on projection
Storage	stationary storage power and energy state	fixed injection in PF/native OPF; parity OPF dispatch variable for AC and DC storage
MobileStorage	relocatable storage resource	projected to AC Storage at current bus
VirtualPowerPlant	aggregated DER portfolio	projected to AC StaticGenerator at aggregation bus
Microgrid	PCC exchange model	projected to AC StaticGenerator at aggregation bus/PCC when grid-connected
DCBus	DC voltage host and DC demand host	direct
DCBranch	DC resistive branch	direct
DCLoad	DC active demand	direct
StaticGeneratorDC	typed DC active-power source with profile/control metadata	stored in <code>dc.dc_static_generators</code> ; serialized and available for DC injection assembly where supported

Struct	Mathematical role	Current steady-state integration mode
PVArrayDC	DC-side photovoltaic array metadata and setpoint	stored in <code>dc.pv_arrays</code> ; serialized and available for DC injection assembly where supported
DCCircuitBreaker	breaker on a DC bus pair	stored and serialized; no projection equivalent currently runs in <code>network_utils.cpp</code>
VSCConverter	AC/DC coupling with loss model and control mode	direct
DCDCConverter	DC/DC transfer device with efficiency and I^2R conduction loss	direct in PF and parity OPF (decision variable with Jacobian and Hessian)
EnergyRouterPort	AC or DC port injection for a multi-port router	direct fixed injection
EnergyRouter	multi-port converter container	direct through its ports
ThreePhaseACBus	optional unbalanced bus model	projected to ACBus
ThreePhaseACLine	optional unbalanced line model	projected to ACBranch using positive sequence
ThreePhaseTransformer	optional unbalanced transformer model	projected to Transformer2W equivalent
ThreePhaseLoad	optional unbalanced load model	projected to Load
ThreePhaseGenerator	optional unbalanced generator model	projected to Generator
ThreePhaseExternalGrid	optional unbalanced slack/grid equivalent	projected to ExternalGrid

8.4 AC Network Elements

ACBus

Field group	Detailed explanation	Used by
Identity and topology	<code>index</code> is the external bus identifier referenced by other components; <code>name</code> is purely descriptive; <code>area</code> and <code>zone</code> are retained for partitioning and reporting.	shared model / solver inputs
Steady-state electrical state	<code>bus_type</code> , <code>vm_pu</code> , <code>va_deg</code> , <code>base_kv</code> , <code>vmax_pu</code> , <code>vmin_pu</code> , <code>in_service</code> . These fields initialize the PF state, define slack/PV/PQ behavior, and bound voltages.	PF, OPF
Embedded demand and shunt	<code>pd_mw</code> , <code>qd_mvar</code> store fallback bus demand when no load table is present; <code>gs_mw</code> , <code>bs_mvar</code> store bus shunt terms.	PF, OPF

Field group	Detailed explanation	Used by
Resilience / geography	n_customers, importance, latitude, longitude. These are not yet decision variables, but they are useful for planning logic, GIS, or load-priority extensions.	planning / future use

ACBranch

Field group	Detailed explanation	Used by
Connectivity	index, from_bus, to_bus, in_service, name. Bus references use external bus IDs.	PF, OPF
Positive-sequence electrical data	r_pu, x_pu, b_pu, tap, shift_deg. These define the AC π -model and branch-flow reporting.	PF, OPF
Ratings and physical metadata	rate_a_mva, rate_b_mva, rate_c_mva, length_k m. rate_a_mva is used directly by OPF thermal limits.	OPF, reporting
Reliability and sequence data	failure_rate, mttr_hr, r0_pu, x0_pu, b0_pu, and transformer-nameplate fields vn_hv_kv, vn_lv_kv, sn_mva. These are stored for planning/fault-model extensions, but no public SC solver is present.	model/I/O meta-data

Transformer2W, Transformer3W, Switch, and CircuitBreaker

Struct	Field-level explanation	Current treatment
Transformer2W	Identity/connectivity fields index, name, hv_bus, lv_bus, in_service ; nameplate fields sn_mva, vn_hv_kv, vn_lv_kv ; short-circuit fields vk_percent, vkr_percent, pk_kw ; no-load fields pf_e_kw, i0_percent ; tap-changer fields tap_side, tap_pos, tap_min, tap_max, tap_neutral, tap_step_percent ; phase shift shift_deg ; zero-sequence fields z0_percent, x0_r0 ; reliability fields mtbf_hours, mttr_hours .	projected to one equivalent ACBranch
Transformer3W	Connectivity hv_bus, mv_bus, lv_bus ; per-winding ratings sn_hv_mva, sn_mv_mva, sn_lv_mva ; pairwise short-circuit data vk_hv_mv_percent, vk_hv_lv_percent, vk_mv_lv_percent and the matching vkr fields; tap data and phase shifts shift_mv_deg, shift_lv_deg .	projected to three equivalent ACBranch objects

Struct	Field-level explanation	Current treatment
Switch	Topology fields <code>bus_from</code> , <code>bus_to</code> , <code>closed</code> ; electrical fields <code>r_contact_ohm</code> , <code>z_ohm</code> , <code>i_rated_ka</code> , <code>i_breaking_ka</code> ; protection fields <code>element_type</code> , <code>element_id</code> ; automation fields <code>is_remote</code> , <code>is_automated</code> , <code>t_operation_s</code> ; reliability fields <code>p_sw_fail</code> , <code>mtbf_hours</code> , <code>mttr_hours</code> , <code>t_scheduled_hr</code> , <code>t_tp_hr</code> .	projected to equivalent branch with <code>in_service=closed</code>
CircuitBreaker	AC breaker fields <code>index</code> , <code>name</code> , <code>bus_from</code> , <code>bus_to</code> , <code>in_service</code> , <code>breaker_type</code> , <code>closed</code> , <code>z_ohm</code> , <code>rated_voltage_kv</code> , <code>i_rated_ka</code> , <code>i_breaking_ka</code> , <code>element_type</code> , <code>element_id</code> .	projected to equivalent branch; if switches or breakers exist, zero-impedance bus merging may run after projection

ExternalGrid and AsynchronousMotor

Struct	Field-level explanation	Current treatment
ExternalGrid	<code>bus</code> , <code>vm_pu</code> , <code>va_deg</code> establish slack voltage targets; <code>s_sc_max_mva</code> , <code>s_sc_min_mva</code> , <code>rx_max</code> , <code>rx_min</code> , <code>r_pu</code> , <code>x_pu</code> , <code>r0_pu</code> , <code>x0_pu</code> , <code>vn_kv</code> , <code>ikq_ka</code> , <code>x_r</code> store source-equivalent/fault metadata; <code>controllable</code> can support dispatch logic.	PF source metadata; additional fields are stored for future analysis
AsynchronousMotor	<code>bus</code> , <code>vn_kv</code> , <code>sn_mva</code> , <code>r_pu</code> , <code>x_pu</code> , <code>x_r</code> , <code>lrc</code> , <code>poles</code> , <code>cos_phi</code> , <code>efficiency</code> , <code>r0_pu</code> , <code>x0_pu</code> . These fields describe possible motor fault contribution, not a steady-state PF state.	model/I/O metadata

8.5 Load Models and Demand-Side Components

Struct	Field-level explanation	Current treatment
Load	Identity fields <code>index</code> , <code>bus</code> , <code>name</code> , <code>in_service</code> ; steady-state demand <code>p_mw</code> , <code>q_mvar</code> ; model selector <code>model</code> ; ZIP coefficients <code>z_percent_p</code> , <code>i_percent_p</code> , <code>p_percent_p</code> , and the matching <code>q</code> fields; demand-response metadata <code>controllable</code> , <code>p_min_mw</code> , <code>cost_mw</code> ; load-priority metadata <code>priority</code> , <code>n_customers</code> , <code>profile_id</code> ; motor-fraction metadata <code>sn_mva</code> , <code>motor_percent</code> , <code>x_sub_pu</code> , <code>r_sc_pu</code> .	direct in PF/OPF

Struct	Field-level explanation	Current treatment
FlexibleLoad	Baseline demand <code>p_mw</code> , <code>q_mvar</code> ; flexibility envelope <code>flex_up_mw</code> , <code>flex_down_mw</code> , <code>flex_duration_h</code> ; response timing <code>response_time_s</code> , <code>ramp_rate_mw_min</code> , <code>availability_pct</code> ; <code>control_area</code> .	projected to equivalent Load for PF; parity OPF has <code>pflex_mw</code> variables
AsymmetricLoad	Per-phase rated values <code>pa_rated_mw</code> , ..., <code>qc_rated_mvar</code> ; per-phase actual values <code>pa_mw</code> , ..., <code>qc_mvar</code> ; connection, grounded, scaling; ZIP shares <code>const_z_percent</code> , <code>const_i_percent</code> , <code>const_p_percent</code> .	projected to aggregated Load
DCLoad	<code>p_mw</code> is the direct active demand at a DC bus; <code>controllable</code> , <code>p_min_mw</code> , <code>cost_mw</code> , and <code>profile_id</code> are control metadata for future extensions.	direct in DC PF and hybrid OPF balances
ChargingStation	<code>bus</code> , <code>n_fast</code> , <code>n_slow</code> , <code>num_chargers</code> ; capacity fields <code>p_fast_max_kw</code> , <code>p_slow_max_kw</code> , <code>max_power_kw</code> ; utilization fields <code>simultaneity_factor</code> , <code>power_factor</code> , <code>utilization_rate</code> ; aggregated operating demand <code>p_total_kw</code> , <code>q_total_kvar</code> .	direct fixed load
Charger	<code>station_id</code> links the charger to its parent station; <code>charger_type</code> , <code>p_rated_kw</code> , <code>p_ch_max_kw</code> , <code>p_ch_min_kw</code> , <code>eta</code> , <code>v2g_capable</code> , <code>p_dis_max_kw</code> .	aggregated into ChargingStation during projection

8.6 Generation, Storage, and Aggregation Components

Struct	Field-level explanation	Current treatment
Generator	Dispatch/state fields <code>pg_mw</code> , <code>qg_mvar</code> , <code>vg_pu</code> , <code>is_slack</code> ; limits <code>pmax_mw</code> , <code>pmin_mw</code> , <code>qmax_mvar</code> , <code>qmin_mvar</code> ; quadratic cost coefficients <code>cost_c2</code> , <code>cost_c1</code> , <code>cost_c0</code> ; unit-commitment metadata <code>startup_cost</code> , <code>shutdown_cost</code> , <code>min_up_time_hr</code> , <code>min_dn_time_hr</code> , <code>ramp_up_mw_min</code> , <code>ramp_dn_mw_min</code> ; machine/stability data <code>mbase_mva</code> , <code>inertia_h</code> , <code>droop_r</code> , <code>xd_pu</code> , <code>xdp_pu</code> , <code>xdpp_pu</code> , <code>td0p_s</code> , <code>td0pp_s</code> ; SC and environmental metadata <code>ra_pu</code> , <code>vn_kv</code> , <code>cos_phi</code> , <code>xq_pu</code> , <code>xd_xq</code> , <code>x0_pu</code> , <code>r0_pu</code> , emission factors, outage rate, and repair time.	direct
StaticGenerator	Fixed or scaled DER injection <code>p_mw</code> , <code>q_mvar</code> , scaling; device rating/limits <code>p_rated_mw</code> , <code>sn_mva</code> , <code>pmax_mw</code> , <code>pmin_mw</code> , <code>qmax_mvar</code> , <code>qmin_mvar</code> ; control metadata <code>controllable</code> , <code>v_ref_pu</code> , <code>k_p</code> , <code>k_q</code> ; SC metadata <code>k</code> , <code>rx</code> .	direct fixed injection

Struct	Field-level explanation	Current treatment
RenewableGen	type, actual output <code>p_mw</code> , <code>q_mvar</code> , rated power <code>p_rated_mw</code> , reactive limits, curtailability flag, curtailment cost, capacity factor, profile pointer, and avoided-emission factor.	fixed injection in PF/native OPF; parity OPF uses <code>pren_mw/qren_mvar</code> for curtailable rows
PVSystem	Actual output <code>p_mw</code> , <code>q_mvar</code> ; inverter limits and control <code>sn_mva</code> , <code>pmax_mw</code> , <code>pmin_mw</code> , <code>qmax_mvar</code> , <code>qmin_mvar</code> , <code>control_mode</code> , <code>controllable</code> , <code>v_ac_set_pu</code> , <code>v_dc_set_pu</code> ; inverter efficiency <code>inverter_eff</code> , <code>loss_percent</code> ; array data <code>num_series</code> , <code>num_parallel</code> , <code>vmpp</code> , <code>impp</code> , <code>voc</code> , <code>isc</code> , <code>alpha_isc</code> , <code>beta_voc</code> ; environmental state <code>irradiance</code> , <code>temperature</code> .	fixed injection in PF/native OPF; parity OPF uses renewable dispatch variables for controllable rows
Storage	Snapshot dispatch <code>p_mw</code> , <code>q_mvar</code> ; power limits <code>p_rated_mw</code> , <code>pmax_mw</code> , <code>pmin_mw</code> , <code>qmax_mvar</code> , <code>qmin_mvar</code> ; energy state <code>e_rated_mwh</code> , <code>soc_init</code> , <code>soc_min</code> , <code>soc_max</code> ; efficiency <code>eta_charge</code> , <code>eta_discharge</code> , <code>self_discharge_pct</code> ; lifecycle <code>max_cycles</code> , <code>current_cycles</code> , <code>soh</code> , <code>replacement_cost</code> .	fixed injection in PF/native OPF; parity OPF has AC and DC storage dispatch variables
MobileStorage	Reuses storage-style power and energy fields and adds mobility fields <code>is_mobile</code> , <code>status</code> , <code>current_location</code> , <code>target_bus</code> , <code>e_consumption_mwh_km</code> , <code>max_travel_distance_km</code> .	projected to AC Storage at its current bus
VirtualPowerPlant	Fleet counts, aggregated capacities, output fields <code>p_output_mw</code> , <code>q_output_mvar</code> , limits, and ramp rates.	projected to AC StaticGenerator at aggregation bus
Microgrid	PCC/aggregation bus, internal bus list, operating-mode fields <code>operating_mode</code> , <code>islanding_capability</code> , <code>auto_reconnection</code> ; exchange limits and current <code>p_exchange_mw</code> ; aggregate capability metrics and droop/voltage setpoints.	projected to AC StaticGenerator when grid-connected

8.7 DC and Power-Electronic Components

Struct	Field-level explanation	Current treatment
DCBus	<code>index</code> , <code>bus_type</code> , <code>vm_pu</code> , <code>vmax_pu</code> , <code>vmin_pu</code> , <code>pd_mw</code> , <code>in_service</code> , <code>name</code> , <code>base_kv</code> , <code>area</code> , <code>zone</code> , <code>n_customers</code> , <code>importance</code> , <code>is_load</code> , <code>latitude</code> , <code>longitude</code> . <code>pd_mw</code> is used only when the dedicated DC load table is empty.	direct

Struct	Field-level explanation	Current treatment
DCBranch	from_bus, to_bus, r_pu, rate_a_mva, length_km, base_kv, s_max_mva, n_parallel, mtbf_hours, mttr_hours, t_scheduled_hr, in_service.	direct
StaticGeneratorDC	index, bus, in_service, name, type, p_set_mw, scaling, profile_id, pmax_mw, pmin_mw, controllable, mtbf_hours, mttr_hours, t_scheduled_hr.	stored in dc.dc_static_generators; serialized as its own DC source family
PVArrayDC	index, bus, in_service, name, p_set_mw, profile_id, array electrical parameters, environmental state, and reliability fields.	stored in dc.pv_arrays; serialized as its own DC PV family
DCCircuitBreaker	index, name, bus_from, bus_to, in_service, breaker_type, closed, r_ohm, rated_voltage_kv, i_rated_ka, i_breaking_ka, element_type, element_id.	stored and serialized; no current projection equivalent
VSCConverter	Connectivity bus_ac, bus_dc; control mode control_mode; topology type; setpoints p_set_mw, q_set_mvar, v_dc_set_pu, v_ac_set_pu; loss parameters eta, loss_percent, loss_mw; control gains k_vdc, k_p, k_q; limits and ratings including vn_ac_kv, vn_dc_kv; controllability, grid-forming and SC-current metadata, and reliability fields.	direct in PF and OPF; some C++ fields are not serialized by the current JSON writer
DCDCConverter	Connectivity bus_in, bus_out; operating mode control_mode; references p_ref_mw (output-side referenced), v_ref_pu; rating fields sn_mva, vn_in_kv, vn_out_kv; efficiency eta; conduction-loss resistance r_eq_pu; droop gain k_droop; limits pmax_mw, pmin_mw; controllable, f_switching_hz, and reliability fields. Sign convention: positive direction is bus_in→bus_out; PF uses output-referenced p_ref_mw, OPF uses input-referenced P^{in} .	direct in PF and parity OPF; some C++ fields are not serialized by the current JSON writer
EnergyRouterPort	bus, port_type, side, voltage_level_kv, operating injection p_mw, q_mvar, v_pu, efficiency eta; limits and control metadata.	expanded by projection into internal converter equivalents
EnergyRouter	Container fields router_type, num_ports, ports; rating/loss fields; control_mode, power_dispatch_strategy, t_scheduled_hr; device-level limits and reliability.	expanded by projection into internal DC buses, VSCs, and DCDC converters

8.8 EnergyRouter Multi-Port Model

An **EnergyRouter** is a power-electronic converter hub that simultaneously connects to an arbitrary number of AC and DC buses. Each physical terminal is represented by one **EnergyRouterPort** entry in the router's **ports** vector.

Port attributes

Field	Type / default	Description
port_type	<code>ERPortType::AC</code>	AC or DC: determines which bus network this port injects into
side	<code>int\{0\}</code>	Physical winding side: 0 = Side A, 1 = Side B (used by SST/UPFC topologies)
bus	<code>int\{0\}</code>	Index of the connected AC or DC bus
p_mw, q_mvar	<code>double\{0\}</code>	Bus-injection sign convention: positive = power into bus
eta	<code>double\{0.98\}</code>	Per-port conversion efficiency for loss allocation
control_mode	<code>ERControlMode::PQ</code>	Per-port control law: PQ (fixed setpoint), VQ (voltage-droop reactive), or Slack (voltage-forming)
p_set_mw, q_set_mvar	<code>double\{0\}</code>	Active/reactive setpoints when control_mode = PQ

Multi-port power balance

Let port k carry active injection P_k (positive = into bus) and efficiency η_k . The lossless internal balance constraint is:

$$\sum_{k=1}^{N_p} P_k = 0 \quad (\text{lossless ER}). \quad (21)$$

When per-port efficiency $\eta_k < 1$, the actual balance allocates losses to each port proportionally:

$$\sum_{k:P_k < 0} |P_k| = \sum_{k:P_k > 0} \frac{P_k}{\eta_k} \quad (\text{lossy ER, simplified allocation}). \quad (22)$$

In the current projection path, an **EnergyRouter** is expanded into a set of internal DC buses, VSC equivalents for each AC port, and DCDC converter equivalents for each DC port. The internal bus forms the power-exchange node; (??) is enforced implicitly by the solver topology.

Router-level fields

Field	Default	Description
router_type	<code>""</code>	Topology identifier, e.g. "SST" (solid-state transformer), "UPFC" (unified power-flow controller)
num_ports	<code>0</code>	Number of EnergyRouterPort entries in ports

Field	Default	Description
control_mode	""	Router-level dispatch strategy: "autonomous" or "centralized"
power_dispatch_strategy	""	Inter-port power allocation policy: "priority" or "proportional"
p_rated_mw	0	Rated through-power capacity of the device
loss_percent	0	Device-level loss as a percentage of rated power

8.9 VirtualPowerPlant Fleet Aggregation Model

A `VirtualPowerPlant` (VPP) represents a portfolio of distributed energy resources (DERs) co-dispatched as a single grid entity. The struct carries both static fleet composition data and dynamic dispatch outputs.

Fleet composition

Field	Default	Meaning
n_pv_systems	0	Number of rooftop/ground-mounted PV units in portfolio
n_wind_turbines	0	Number of wind turbines
n_battery_systems	0	Number of battery storage units
n_ev_chargers	0	Number of V2G-capable EV chargers
n_controllable_loads	0	Demand-response resources
n_chp	0	Combined heat-and-power units
n_biomass, n_thermal_storage, n_hvac, n_industrial	0	Other flexible DER categories
p_pv_sum_mw, p_wind_sum_mw	0	Aggregated renewable capacity
e_storage_sum_mwh	0	Total usable storage energy

Aggregation dispatch model

Let P_i^{DER} be the net active output of DER unit i (positive = generation, negative = consumption). The VPP aggregates to:

$$P^{\text{VPP}} = \sum_{i \in \mathcal{F}} P_i^{\text{DER}} = P^{\text{gen}} - P^{\text{flex,abs}}, \quad (23)$$

where P^{gen} is the total scheduled generation and $P^{\text{flex,abs}}$ is the absorbed flexible load. The field `p_output_mw` stores this aggregated value; `q_output_mvar` carries the corresponding reactive output.

Regulation capability is bounded by:

$$-P^{\text{reg,down}} \leq \Delta P \leq P^{\text{reg,up}}, \quad (24)$$

stored in `p_regulation_down_mw` and `p_regulation_up_mw`.

Projection to grid model

In the current solver path, a `VirtualPowerPlant` is projected to a single `AC StaticGenerator` at `aggregation_bus` with:

$$p_{sg} = p_output_mw, \quad q_{sg} = q_output_mvar.$$

Ramp-rate fields (`ramp_up_max_mw_min`, `ramp_down_max_mw_min`) and regulation bounds are available to time-series and OPF formulations but are not yet enforced by the snapshot PF solver.

8.10 Optional Three-Phase Models

Struct	Field-level explanation	Current treatment
<code>ThreePhaseACBus</code>	Per-phase voltages <code>vm_a_pu</code> , <code>va_a_deg</code> , <code>vm_b_pu</code> , ...; per-phase demand and shunts; common voltage limits, area, and zone.	projected to <code>ACBus</code> by phase summation / averaging
<code>ThreePhaseACLine</code>	Positive- and zero-sequence per-km parameters, optional precomputed per-unit values, parallel count, thermal rating, and reliability data.	projected to <code>AC Branch</code> using positive-sequence values
<code>ThreePhaseTransformer</code>	Two-winding transformer with vector group, positive- and zero-sequence short-circuit data, tap changer, and phase shift.	projected to <code>Transformer2W</code> equivalent before branch projection
<code>ThreePhaseLoad</code>	Connection type, grounding flag, per-phase powers, ZIP shares, and motor fraction.	projected to aggregated <code>Load</code>
<code>ThreePhaseGenerator</code>	Per-device dispatch and limits plus sequence impedances <code>xd_pu</code> , <code>xdpp_pu</code> , <code>x2_pu</code> , <code>x0_pu</code> , <code>r0_pu</code> .	projected to <code>Generator</code>
<code>ThreePhaseExternalGrid</code>	Per-sequence grid-equivalent impedances and short-circuit capacities.	projected to <code>ExternalGrid</code>
<code>ThreePhaseBusResult</code>	Per-phase solved voltage and injection report. The current one-line notebook path documents the model, but the main steady-state solvers still operate on projected single-phase equivalents.	reporting / future unbalanced workflows

9 I/O Contracts, Sheet Dictionaries, and JSON Keys

The model dictionaries in the previous section explain the physical meaning of each field. This section documents the external I/O contract actually exposed by the current C++ implementation: JSON path names, Excel sheet names, column names, defaulting rules, aggregated views, and the current serialization gaps.

9.1 Serialization Families

API entry point	Artifact	Direction	Practical behavior
<code>to_json / from_json</code>	one JSON document for <code>HybridPowerSystem</code>	bidirectional	covers the major public model families attached to <code>HybridPowerSystem</code> , including circuit breakers, DC static generators, DC PV arrays, DC circuit breakers, <code>ac.motors</code> , and optional <code>three_phase_ac</code> ; several C++ struct fields are still not serialized.
<code>load_excel / save_excel</code>	public Excel API declared in <code>excel_io.hpp</code>	bidirectional when enabled	without <code>HACDCPF_ENABLE_EXCEL</code> , both inline functions throw; with Excel enabled, the header declares these names, while the current source defines the workbook helpers below under <code>load_xlsx/save_xlsx</code> .
source-only <code>save_xlsx / load_xlsx</code>	engineering workbook with one table per supported component class	bidirectional when compiled	implemented in <code>src/io/excel_io.cpp</code> ; input-oriented Excel template plus derived aggregated views such as <code>ACLoad</code> and <code>DCLoad</code> .
source-only PF workbook helpers	dedicated/compact PF result workbooks	export only	<code>src/io/excel_io.cpp</code> implements result workbook writers, but these helper names are not declared in <code>include/hacdcpf/io/excel_io.hpp</code> .
source-only OPF workbook helpers	dedicated/compact OPF result workbooks	export only	write <code>OPF_*</code> sheets; compact OPF export does not append <code>result_*</code> columns into input tables.
<code>power_flow_result_to_json / save_power_flow_result_json</code>	PF result JSON	export only	serializes convergence summary plus AC/DC bus states, AC/DC branch flows, VSC transfers, and DCDC transfers.
<code>opf_result_to_json / opf_result_from_json</code>	AC OPF result JSON	bidirectional	serializes solver status, solver path, primal solution vectors, enhanced dispatch arrays with identity maps, legacy dispatch arrays, and profiling summary.

API entry point	Artifact	Direction	Practical behavior
<code>save_carbon_results_xlsx / carbon_result_to_json</code>	carbon-analysis workbook / JSON	export only for Excel, bidirectional for JSON	uses dedicated report-oriented structures, not the main system model.

9.2 Global Parsing and Workbook Rules

Rule	Current implementation behavior
JSON missing keys	Most JSON importers use <code>jget(..., default)</code> ; missing optional keys are silently defaulted. Whole arrays are optional as well.
JSON required keys	Identity and connectivity fields are usually mandatory: for example <code>index</code> , <code>bus</code> , <code>from_bus</code> , <code>to_bus</code> , <code>hv_bus</code> , <code>lv_bus</code> , <code>bus_ac</code> , and <code>bus_dc</code> .
JSON enums	Enum values are string-coded: for example <code>PQ/PV/SLACK</code> , <code>DC_V/DC_P</code> , <code>VDC_Q</code> , <code>Wind</code> , <code>FuelCell</code> , <code>Recloser</code> , <code>Islanded</code> .
Excel header row	All Excel tables use row 1 as the header row. Data rows start at row 2.
Excel row termination	Readers stop at the first row with an empty <code>index</code> cell. This means blank rows inside a table terminate import early.
Excel missing sheets	<code>load_xlsx</code> silently skips missing sheets; the corresponding component vector remains empty.
Excel boolean parser	<code>TRUE</code> , <code>T</code> , <code>YES</code> , <code>Y</code> , and <code>1</code> map to <code>true</code> ; <code>FALSE</code> , <code>F</code> , <code>NO</code> , <code>N</code> , and <code>0</code> map to <code>false</code> .
Sheet precedence	For AC loads, <code>Load</code> is preferred; <code>ACLoad</code> is used only as a fallback. For DC loads, <code>DCLoadRaw</code> is preferred; <code>DCLoad</code> is only a fallback.
Bus-demand synchronization	After Excel import, <code>sync_bus_demands_from_load_tables(sys)</code> is called so the bus-level embedded demand reflects the explicit load tables.
Aggregated Excel views	<code>ACBus</code> and <code>DCBus</code> written by the Excel exporters are aggregated bus views, not always the raw struct contents; <code>ACLoad</code> and <code>DCLoad</code> are explicitly derived tables.
Empty result sheets in templates	<code>save_xlsx</code> currently pre-creates the <code>Results_*</code> sheets even though it does not fill them. A freshly exported template therefore contains empty result tabs.

9.3 Hybrid System JSON Contract

Top-level JSON object

JSON path	Value type	Current meaning
<code>name</code>	scalar string	top-level <code>HybridPowerSystem::name</code> .
<code>base_mva</code>	scalar number	top-level system base.

JSON path	Value type	Current meaning
<code>ac</code>	object	AC subsystem object with its own name, base, frequency, and AC component arrays.
<code>dc</code>	object	DC subsystem object with its own name, base, and DC component arrays.
<code>vsc_converters</code>	array	top-level VSC array.
<code>dc.dcdc_converters</code>	array	exported DC/DC converter array nested inside the DC subsystem; import also accepts legacy top-level <code>dcdc_converters</code> .
<code>energy_routers</code>	array	top-level multi-port router array; ports are nested inside each router object.
<code>mobile_storage</code>	array	top-level mobile-storage array.
<code>vpps</code>	array	top-level virtual power plant array.
<code>microgrids</code>	array	top-level microgrid array.
<code>three_phase_ac</code>	object	optional unbalanced AC subsystem with its own base, frequency, and three-phase component arrays.

Current JSON caveats

Public model area	Current JSON status
Standalone energy-router port array	not used; ports appear only as nested objects under each <code>energy_routers[i].ports</code> .
Load short-circuit sub-fields	the embedded SC motor-fraction fields <code>sn_mva</code> , <code>motor_percent</code> , <code>x_sub_pu</code> , and <code>r_sc_pu</code> are still not part of the main <code>ac.loads[]</code> JSON contract.
Excel API names	the public header exposes <code>load_excel/save_excel</code> , but the optional source file currently defines <code>load_xlsx/save_xlsx</code> and additional workbook result helpers that are not public-header declarations.
DC bus/branch struct fields	<code>DCBus</code> and <code>DCBranch</code> have more C++ struct fields than the current JSON writer exports; see Section 02b_io_field_dictionaries.

AC-network and demand JSON objects

Model	JSON path	Required keys on import	Other exported keys
ACBus	<code>ac.buses[]</code>	<code>index</code>	<code>bus_type</code> , <code>pd_mw</code> , <code>qd_mvar</code> , <code>vm_pu</code> , <code>va_deg</code> , <code>area</code> , <code>base_kv</code> , <code>vmax_pu</code> , <code>vmin_pu</code> , <code>gs_mw</code> , <code>bs_mvar</code> , <code>zone</code> , <code>in_service</code> , <code>name</code> , <code>n_customers</code> , <code>importance</code> , <code>latitude</code> , <code>longitude</code>

Model	JSON path	Required keys on import	Other exported keys
ACBranch	ac.branches[]	index, from_bus, to_bus	r_pu, x_pu, b_pu, tap, shift_deg, rate_a_mva, rate_b_mva, rate_c_mva, in_service, name, length_km, failure_rate, mttr_hr
Load	ac.loads[]	index, bus	in_service, name, p_mw, q_mvar, model, z_percent_p, i_percent_p, p_percent_p, z_percent_q, i_percent_q, p_percent_q, controllable, p_min_mw, cost_mw, priority, n_customers, profile_id
FlexibleLoad	ac.flexible_loads[]	index, bus	in_service, name, p_mw, q_mvar, flex_up_mw, flex_down_mw, flex_duration_h, response_time_s, ramp_rate_mw_min, availability_pct, controllable, priority, control_area
AsymmetricLoad	ac.asymmetric_loads[]	index, bus	in_service, name, connection, grounded, pa_rated_mw, qa_rated_mvar, pb_rated_mw, qb_rated_mvar, pc_rated_mw, qc_rated_mvar, pa_mw, qa_mvar, pb_mw, qb_mvar, pc_mw, qc_mvar, scaling, const_z_percent, const_i_percent, const_p_percent, controllable, priority
Shunt	ac.shunts[]	index, bus	in_service, name, gs_mw, bs_mvar, switchable, n_steps, current_step, bs_per_step
Switch	ac.switches[]	index, bus_from, bus_to	name, in_service, switch_type, closed, r_contact_ohm, z_ohm, i_rated_ka, i_breaking_ka, element_type, element_id, is_remote, is_automated, t_operation_s, p_sw_fail, mtbf_hours, mttr_hours
CircuitBreaker	ac.circuit_breakers[]	index, bus_from, bus_to	name, in_service, breaker_type, closed, z_ohm, rated_voltage_kv, i_rated_ka, i_breaking_ka, element_type, element_id
ChargingStation	ac.charging_stations[]	index, bus	name, location, in_service, n_fast, n_slow, num_chargers, p_fast_max_kw, p_slow_max_kw, max_power_kw, simultaneity_factor, power_factor, utilization_rate, p_total_kw, q_total_kvar, mtbf_hours, mttr_hours
Charger	ac.chargers[]	index	name, station_id, charger_type, in_service, p_rated_kw, p_ch_max_kw, p_ch_min_kw, eta, v2g_capable, p_dis_max_kw, mtbf_hours, mttr_hours
Asynchronous Motor	ac.motors[]	index, bus	name, in_service, machine ratings, direct / zero-sequence parameters, locked-rotor data, and efficiency metadata
ExternalGrid	ac.external_grids[]	index, bus	name, in_service, vm_pu, va_deg, s_sc_max_mva, s_sc_min_mva, rx_max, rx_min, r_pu, x_pu, r0_pu, x0_pu, vn_kv, controllable

Model	JSON path	Required keys on import	Other exported keys
Transformer2W	ac.transformers_2w[]	index, hv_bus, lv_bus	name, std_type, in_service, sn_mva, vn_hv_kv, vn_lv_kv, vk_percent, vkr_percent, pk_kw, pfe_kw, i0_percent, tap_side, tap_pos, tap_min, tap_max, tap_neutral, tap_step_percent, shift_deg, vector_group, z0_percent, x0_r0, mtbf_hours, mttr_hours
Transformer3W	ac.transformers_3w[]	index, hv_bus, mv_bus, lv_bus	name, std_type, in_service, sn_hv_mva, sn_mv_mva, sn_lv_mva, vn_hv_kv, vn_mv_kv, vn_lv_kv, vk_hv_mv_percent, vk_hv_lv_percent, vk_mv_lv_percent, vkr_hv_mv_percent, vkr_hv_lv_percent, vkr_mv_lv_percent, pfe_kw, i0_percent, tap_side, tap_pos, tap_step_percent, shift_mv_deg, shift_lv_deg, mtbf_hours, mttr_hours

Generation, storage, and hybrid JSON objects

Model	JSON path	Required keys on import	Other exported keys
Generator	ac.generators[]	index, bus	in_service, pg_mw, qg_mvar, vg_pu, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, is_slack, cost_c2, cost_c1, cost_c0, name, fuel_type, startup_cost, shutdown_cost, min_up_time_hr, min_dn_time_hr, ramp_up_mw_min, ramp_dn_mw_min, mbase_mva, inertia_h, droop_r, xd_pu, xdp_pu, xdpp_pu, td0p_s, td0pp_s, emission_factor_tco2_mwh, nox_factor_kg_mwh, so2_factor_kg_mwh, forced_outage_rate, mttr_hr
StaticGenerator	ac.static_generators[] or dc.static_generators[]	index, bus	in_service, name, sgen_type, p_mw, q_mvar, p_rated_mw, sn_mva, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, scaling, controllable, v_ref_pu, k_p, k_q, k, rx, co2_emission_rate, mtbf_hours, mttr_hours
RenewableGen	ac.renewable_generators[]	index, bus	in_service, name, type, p_mw, q_mvar, p_rated_mw, qmax_mvar, qmin_mvar, curtailable, cost_curtail_mwh, capacity_factor, profile_id, emission_offset_tco2_mwh

Model	JSON path	Required keys on import	Other exported keys
PVSystem	ac.pv_systems[]	index, bus	in_service, name, p_mw, q_mvar, sn_mva, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, control_mode, controllable, v_ac_set_pu, v_dc_set_pu, inverter_eff, loss_percent, num_series, num_parallel, vmpp, impp, voc, isc, alpha_isc, beta_voc, irradiance, temperature, mtbf_panel_hours, mttr_panel_hours, mtbf_inverter_hours, mttr_inverter_hours
Storage	ac.storage[] or dc.storage[]	index, bus	in_service, name, p_mw, q_mvar, p_rated_mw, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, e_rated_mwh, soc_init, soc_min, soc_max, soc_carbon_intensity_tco2_mwh, eta_charge, eta_discharge, self_discharge_pct, max_cycles, current_cycles, soh, replacement_cost, forced_outage_rate, mttr_hr
DCBus	dc.buses[]	index	bus_type, vm_pu, vmax_pu, vmin_pu, pd_mw, in_service, name, latitude, longitude
DCBranch	dc.branches[]	index, from_bus, to_bus	r_pu, rate_a_mva, in_service, name, length_km
VSCConverter	vsc_converters[]	index, bus_ac, bus_dc	in_service, control_mode, p_set_mw, q_set_mvar, v_dc_set_pu, v_ac_set_pu, eta, loss_percent, loss_mw, k_vdc, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, p_rated_mw, name, forced_outage_rate, mttr_hr
StaticGeneratorDC	dc.dc_static_generators[]	index, bus	in_service, name, type, p_set_mw, scaling, profile_id, pmax_mw, pmin_mw, controllable, mtbf_hours, mttr_hours, t_scheduled_hr
PVArrayDC	dc.pv_arrays[]	index, bus	in_service, name, p_set_mw, profile_id, num_series, num_parallel, vmpp, impp, voc, isc, alpha_isc, beta_voc, temperature, irradiance, mtbf_hours, mttr_hours, t_scheduled_hr
DCCircuitBreaker	dc.dc_circuit_breakers[]	index, bus_from, bus_to	name, in_service, breaker_type, closed, r_ohm, rated_voltage_kv, i_rated_ka, i_breaking_ka, element_type, element_id
DCDCConverter	dc.dcdc_converters[]	index, bus_in, bus_out	in_service, name, control_mode, p_ref_mw, v_ref_pu, sn_mva, vn_in_kv, vn_out_kv, eta, r_eq_pu, pmax_mw, pmin_mw, k_droop, mtbf_hours, mttr_hours
EnergyRouterPort	energy_routers[].ports[]	index, bus	name, port_type, voltage_level_kv, p_mw, q_mvar, v_pu, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, control_mode, p_set_mw, q_set_mvar, v_set_pu, in_service

Model	JSON path	Required keys on import	Other exported keys
EnergyRouter	energy_routers[]	index	name, in_service, router_type, num_ports, ports, p_rated_mw, vn_ac_kv, vn_dc_kv, loss_percent, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, mtbf_hours, mttr_hours
MobileStorage	mobile_storage[]	index, bus	in_service, name, p_mw, q_mvar, p_rated_mw, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, e_rated_mwh, soc_init, soc_min, soc_max, eta_charge, eta_discharge, max_cycles, current_cycles, soh, is_mobile, status, current_location, target_bus, e_consumption_mwh_km, max_travel_distance_km, mtbf_hours, mttr_hours
VirtualPower Plant	vpps[]	index	name, description, pcc_bus, resource-count fields, aggregate capability / output terms, ramp limits, reliability metadata, and in_service; import also accepts legacy aggregation_bus
Microgrid	microgrids[]	index	name, description, in_service, pcc_bus, internal_buses, operating_mode, islanding_capability, auto_reconnection, p_exchange_max_mw, p_exchange_min_mw, p_import_max_mw, p_export_max_mw, p_exchange_mw, total_generation_mw, total_storage_mwh, total_load_mw, capacity_mw, peak_load_mw, f_set_hz, v_set_pu, k_droop, area, mtbf_hours, mttr_hours
ThreePhase ACSystem	three_phase_ac	none	name, base_mva, base_freq_hz, and component arrays for buses, lines, transformers, loads, generators, and external grids
ThreePhaseAC Bus	three_phase_ac.buses[]	index	bus_type, name, base_kv, in_service, vm_a_pu, va_a_deg, vm_b_pu, va_b_deg, vm_c_pu, va_c_deg, pd_a_mw, qd_a_mvar, pd_b_mw, qd_b_mvar, pd_c_mw, qd_c_mvar, gs_a_mw, bs_a_mvar, gs_b_mw, bs_b_mvar, gs_c_mw, bs_c_mvar, vmin_pu, vmax_pu, area, zone
ThreePhaseAC Line	three_phase_ac.lines[]	index, from_bus, to_bus	name, in_service, length_km, parallel, r1_ohm_per_km, x1_ohm_per_km, c1_nf_per_km, r0_ohm_per_km, x0_ohm_per_km, c0_nf_per_km, r1_pu, x1_pu, b1_pu, r0_pu, x0_pu, b0_pu, max_i_ka, rate_a_mva, failure_rate, mttr_hr
ThreePhase Transformer	three_phase_ac.transformers[]	index, hv_bus, lv_bus	name, in_service, nameplate data, leakage and magnetizing terms, zero-sequence data, tap settings, vector group, and reliability metadata

Model	JSON path	Required keys on import	Other exported keys
ThreePhaseLoad	three_phase_ac.loads[]	index, bus	name, in_service, connection, grounded, p_a_mw, q_a_mvar, p_b_mw, q_b_mvar, p_c_mw, q_c_mvar, const_z_percent, const_i_percent, const_p_percent, motor_percent, lrc_pu, x_r_ratio
ThreePhaseGenerator	three_phase_ac.generators[]	index, bus	name, in_service, is_slack, dispatch setpoints / limits, voltage target, machine base, and sequence reactance terms
ThreePhaseExternalGrid	three_phase_ac.external_grids[]	index, bus	name, in_service, voltage target, short-circuit level bounds, R/X bounds, and sequence Thevenin terms

Result JSON objects

JSON artifact	Top-level keys	Current note
PowerFlowResult JSON	converged, iterations, residual, ac_buses, dc_buses, ac_branch_flows, dc_branch_flows, vsc_transfers, dcdc_transfers	export-only helper writes solved bus states, AC/DC branch flow and loading fields, and converter transfer reports.
ACOPFResult JSON	converged, iterations, outer_iterations, objective, max_constraint_violation, max_stationarity, status, solver_path, vm, va, vdc, pg_mw, qg_mvar, dpd_mw, dqd_mvar, pac_mw, qac_mvar, enhanced dispatch objects, legacy dispatch arrays, profiling	enhanced dispatch objects are renewable_dispatch, storage_dispatch, dcdc_dispatch, and flex_dispatch; each contains values plus a map of original_index/source_type.
CarbonAnalysisResult JSON	tracing_verified, matrix_solved, matrix_residual, load_carbon, dc_load_carbon, branch_carbon, dc_branch_carbon, bus_carbon, dc_bus_carbon, vsc_carbon, dcdc_carbon, storage_carbon, energy_router_carbon, generator_loss_allocation, tracing_summary, matrix_summary	AC/DC load arrays carry the same scalar fields plus nested generator_supply_mw; branch / converter / router arrays expose loss and direction fields; storage_carbon adds charge/discharge-side storage semantics including traced charging intensity.

9.4 Excel Input Workbook Contract

Input-sheet catalog

Sheet	Primary model / load target	Columns	Current note
Info	global metadata	Key, Value	loader recognizes name, base_mva, ac_freq_hz, three_ phase_name, three_ phase_base_mva, and three_phase_ base_freq_hz.
ACBus	ACBus	index, name, bus_type, pd_mw, qd_mvar, vm_pu, va_deg, base_kv, vmin_pu, vmax_pu, gs_mw, bs_mvar, area, zone, in_service, n_customers, importance, latitude, longitude	write path uses aggregated AC demand, not necessarily the raw bus fields.
ACBranch	ACBranch	index, name, from_bus, to_bus, r_pu, x_pu, b_pu, tap, shift_deg, rate_a_mva, rate_b_mva, rate_c_mva, length_km, in_service, failure_rate, mttr_hr	near-1:1 with JSON coverage.
Generator	Generator	index, name, bus, pg_mw, qg_mvar, vg_pu, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, is_slack, fuel_type, mbase_mva, cost_c2, cost_c1, cost_c0, in_service, inertia_h, droop_r, ramp_up_mw_min, ramp_dn_mw_min	Excel omits the deeper machine, emissions, and commitment fields that JSON keeps.
Load	Load	index, name, bus, p_mw, q_mvar, model, z_percent_p, i_percent_p, p_percent_p, z_percent_q, i_percent_q, p_percent_q, controllable, priority, cost_mw, in_service	raw explicit AC load table.
ACLoad	fallback Load import; aggregated export view	same columns as Load	written as a derived table that aggregates Load, charging stations, charging storage, VPP imports, microgrid imports, and AC router ports.
FlexibleLoad	FlexibleLoad	index, name, bus, p_mw, q_mvar, flex_up_mw, flex_down_mw, flex_duration_h, response_time_s, ramp_rate_mw_min, availability_pct, controllable, priority, control_area, in_service	dedicated Excel input sheet now exists; these rows are converted to canonical Load equivalents before PF and also have parity OPF variables.
AsymmetricLoad	AsymmetricLoad	index, name, bus, connection, grounded, paRated_mw, qaRated_mvar, pbRated_mw, qbRated_mvar, pcRated_mw, qcRated_mvar, pa_mw, qa_mvar, pb_mw, qb_mvar, pc_mw, qc_mvar, scaling, const_z_percent, const_i_percent, const_p_percent, controllable, priority, in_service	dedicated Excel input sheet now exists; solver-side use still relies on projection to a single Load equivalent.
StaticGenerator	AC StaticGenerator	index, name, bus, sgen_type, p_mw, q_mvar, pRated_mw, sn_mva, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, scaling, controllable, in_service	Excel omits control, SC, emissions, and reliability meta- data.
Storage	AC Storage	index, name, bus, p_mw, q_mvar, pRated_mw, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, eRated_mwh, soc_init, soc_min, soc_max, eta_charge, eta_discharge, in_service	Excel omits life- cycle and outage fields.

Sheet	Primary model / load target	Columns	Current note
RenewableGen	RenewableGen	index, name, bus, type, p_mw, q_mvar, p_rate_d_mw, qmax_mvar, qmin_mvar, curtailable, capacity_factor, in_service	Excel omits cost_curtail_mwh, profile_id, and emission-offset metadata.
PVSystem	PVSystem	index, name, bus, p_mw, q_mvar, sn_mva, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, control_mode, inverter_eff, in_service	Excel keeps only a narrow operational subset; JSON is much richer.
Shunt	Shunt	index, name, bus, gs_mw, bs_mvar, switchable, n_steps, current_step, bs_per_step, in_service	near-1:1 with JSON coverage.
ExternalGrid	ExternalGrid	index, name, bus, vm_pu, va_deg, s_sc_max_mva, s_sc_min_mva, rx_max, rx_min, controllable, in_service	Excel omits explicit Thevenin impedance fields present in JSON.
Transformer2W	Transformer2W	index, name, hv_bus, lv_bus, sn_mva, vn_hv_kv, vn_lv_kv, vk_percent, vkr_percent, pfe_kw, i0_percent, tap_side, tap_pos, tap_min, tap_max, tap_neutral, tap_step_percent, shift_deg, vector_group, in_service	Excel omits std_type, pk_kw, zero-sequence, and reliability fields.
Transformer3W	Transformer3W	index, name, std_type, hv_bus, mv_bus, lv_bus, sn_hv_mva, sn_mv_mva, sn_lv_mva, vn_hv_kv, vn_mv_kv, vn_lv_kv, vk_hv_mv_percent, vk_hv_lv_percent, vk_mv_lv_percent, vkr_hv_mv_percent, vkr_hv_lv_percent, vkr_mv_lv_percent, pfe_kw, i0_percent, tap_side, tap_pos, tap_step_percent, shift_mv_deg, shift_lv_deg, mtbf_hours, mttr_hours, in_service	dedicated Excel input sheet now exists; solver-side treatment still projects the device into equivalent branches.
ChargingStation	ChargingStation	index, name, bus, n_fast, n_slow, num_chargers, p_fast_max_kw, p_slow_max_kw, max_power_kw, simultaneity_factor, power_factor, p_total_kw, q_total_kvar, in_service	Excel omits location, utilization_rate, and reliability fields.
Charger	Charger	index, name, station_id, charger_type, in_service, p_rated_kw, p_ch_max_kw, p_ch_min_kw, eta, v2g_capable, p_dis_max_kw, mtbf_hours, mttr_hours	dedicated Excel input sheet now exists; charger rows are linked to their parent station by station_id.
Asynchronous Motor	Asynchronous Motor	index, name, bus, machine ratings, direct / zero-sequence parameters, SC motor data, and in_service	dedicated Excel input sheet now exists for short-circuit motor data.
Switch	Switch	index, name, bus_from, bus_to, switch_type, closed, in_service	Excel keeps only the topology/status subset.
CircuitBreaker	CircuitBreaker	index, name, bus_from, bus_to, breaker_type, closed, z_ohm, rated_voltage_kv, i_rated_ka, i_breaking_ka, element_type, element_id, in_service	dedicated AC breaker sheet.
DCBus	DCBus	index, name, bus_type, vm_pu, vmax_pu, vmin_pu, pd_mw, in_service	write path uses aggregated DC demand.
DCLoad	fallback DCLoad import; aggregated export view	index, name, bus, p_mw, controllable, p_min_mw, cost_mw, in_service	written as derived aggregated DC demand.

Sheet	Primary model / load target	Columns	Current note
DCLoadRaw	raw DCLoad table	same columns as DCLoad	preserves the unaggregated DC load objects and is preferred on import.
DCBranch	DCBranch	index, name, from_bus, to_bus, r_pu, rate_a_mva, length_km, in_service	near-1:1 with JSON coverage.
DCStorage	DC Storage	same columns as Storage	same coverage limits as AC storage.
DCStaticGen	DC StaticGenerator	same columns as StaticGenerator	same coverage limits as AC static generators.
DCStaticGenerator	StaticGeneratorDC	index, name, bus, type, p_set_mw, scaling, profile_id, pmax_mw, pmin_mw, controllable, mtbf_hr, mttr_hr, t_scheduled_hr, in_service	preferred explicit typed DC source sheet; importer also accepts legacy DCStaticGeneratorDC.
DCPVArray	PVArrayDC	index, name, bus, p_set_mw, profile_id, num_series, num_parallel, vmpp, impp, voc, isc, alpha_isc, beta_voc, temperature, irradiance, mtbf_hr, mttr_hr, t_scheduled_hr, in_service	typed DC PV array sheet.
DCCircuitBreaker	DCCircuitBreaker	index, name, bus_from, bus_to, breaker_type, closed, r_ohm, rated_voltage_kv, i_rated_ka, i_breaking_ka, element_type, element_id, in_service	dedicated DC breaker sheet.
VSCConverter	VSCConverter	index, name, bus_ac, bus_dc, control_mode, p_set_mw, q_set_mvar, v_dc_set_pu, v_ac_set_pu, eta, loss_percent, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, p_rated_mw, in_service	Excel omits loss_mw, k_vdc, and outage metadata.
DCDCConverter	DCDCConverter	index, name, bus_in, bus_out, control_mode, p_ref_mw, v_ref_pu, sn_mva, eta, pmax_mw, pmin_mw, in_service	Excel omits voltage-rating, droop, resistance, and reliability fields.
EnergyRouter	EnergyRouter	index, name, router_type, num_ports, p_rated_mw, vn_ac_kv, vn_dc_kv, loss_percent, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, mtbf_hours, mttr_hours, in_service	router-level metadata now has a dedicated Excel sheet.
EnergyRouterPort	EnergyRouterPort	router_index, index, name, bus, port_type, voltage_level_kv, p_mw, q_mvar, v_pu, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, control_mode, p_set_mw, q_set_mvar, v_set_pu, in_service	port rows are matched back to their parent router through router_index.
VPP	VirtualPowerPlant	index, name, pcc_bus, aggregate output / limit snapshot, selected resource counts, and in_service	Excel writes pcc_bus from aggregation_bus; it omits description, EV/load counts, aggregate capability, ramp rates, and reliability.
Microgrid	Microgrid	index, name, pcc_bus, operating_mode, islanding_capability, p_exchange_mw, p_exchange_max_mw, total_generation_mw, total_load_mw, in_service	Excel omits internal_buses and most planning/control metadata.

Sheet	Primary model / load target	Columns	Current note
MobileStorage	MobileStorage	index, name, bus, status, p_mw, q_mvar, p_rated_mw, e_rated_mwh, soc_init, in_service	Excel keeps only a snapshot subset and omits most mobility/limit metadata.
ThreePhaseBus	ThreePhaseAC Bus	index, name, bus_type, base_kv, vm_a_pu, va_a_deg, vm_b_pu, va_b_deg, vm_c_pu, va_c_deg, pd_a_mw, qd_a_mvar, pd_b_mw, qd_b_mvar, pd_c_mw, qd_c_mvar, gs_a_mw, bs_a_mvar, gs_b_mw, bs_b_mvar, gs_c_mw, bs_c_mvar, vmin_pu, vmax_pu, area, zone, in_service	explicit unbalanced bus table for the optional three_phase_ac subsystem.
ThreePhaseLine	ThreePhaseAC Line	index, name, from_bus, to_bus, length_km, parallel, r1_ohm_per_km, x1_ohm_per_km, c1_nf_per_km, r0_ohm_per_km, x0_ohm_per_km, c0_nf_per_km, r1_pu, x1_pu, b1_pu, r0_pu, x0_pu, b0_pu, max_i_ka, rate_a_mva, failure_rate, mttr_hr, in_service	sequence-parameter line sheet for explicit unbalanced input.
ThreePhaseTransformer	ThreePhaseTransformer	index, name, buses, nameplate data, leakage / zero-sequence terms, tap settings, reliability metadata, and in_service	explicit three-phase transformer input sheet; solver path still projects it to Transformer2W .
ThreePhaseLoad	ThreePhaseLoad	index, name, bus, connection, grounded, p_a_mw, q_a_mvar, p_b_mw, q_b_mvar, p_c_mw, q_c_mvar, const_z_percent, const_i_percent, const_p_percent, motor_percent, lrc_pu, x_r_ratio, in_service	explicit unbalanced demand table; projected before main PF/OPF assembly.
ThreePhaseGenerator	ThreePhaseGenerator	index, name, bus, slack flag, dispatch setpoints / limits, sequence reactance terms, and in_service	explicit unbalanced generator sheet with sequence impedance metadata.
ThreePhaseExternalGrid	ThreePhaseExternalGrid	index, name, bus, voltage target, SC level bounds, sequence impedance terms, and in_service	explicit three-phase grid-equivalent sheet.

Remaining field-level Excel caveats

Area	Current Excel status
Load short-circuit motor subfields	the embedded SC fields sn_mva , motor_percent , x_sub_pu , and r_sc_pu still do not have dedicated Load/ACLoad workbook columns.
ThreePhaseACSystem metadata container	subsystem metadata is carried in Info keys rather than a dedicated ThreePhaseInfo sheet.

JSON Key / Excel Column Dictionary for the Newly Added Contracts

The next tables are intentionally 1:1 with the current serializer code. The “JSON default” text below means the value used by `from_json` when a key is absent. Excel uses the exact sheet and column names shown here.

Optional three-phase subsystem metadata: `three_phase_ac` and `Info`

Field	Excel sheet / column	JSON default / import rule	Meaning
<code>name</code>	<code>Info[three_phase_name]</code>	default "Three-Phase ACSystem"	user-visible subsystem name for the optional explicit unbalanced AC network
<code>base_mva</code>	<code>Info[three_phase_base_mva]</code>	default 100.0	per-subsystem power base; independent from top-level <code>base_mva</code>
<code>extttCircuitBreaker</code>	<code>ac.circuit_breakers[]</code>	<code>index</code> , <code>bus_from</code> , <code>bus_to</code>	<code>name</code> , <code>in_service</code> , <code>breaker_type</code> , <code>closed</code> , <code>z_ohm</code> , <code>rated_voltage_kv</code> , <code>i_rated_ka</code> , <code>i_breaking_ka</code> , <code>element_type</code> , <code>element_id</code>
<code>base_freq_hz</code>	<code>Info[three_phase_base_freq_hz]</code>	default 50.0	nominal subsystem frequency in Hz
<code>buses[]</code>	sheet <code>ThreePhaseBus</code>	missing array / sheet keeps vector empty	explicit three-phase bus family
<code>lines[]</code>	sheet <code>ThreePhaseLine</code>	missing array / sheet keeps vector empty	explicit three-phase line family
<code>transformers[]</code>	sheet <code>ThreePhaseTransformer</code>	missing array / sheet keeps vector empty	explicit three-phase transformer family
<code>loads[]</code>	sheet <code>ThreePhaseLoad</code>	missing array / sheet keeps vector empty	explicit unbalanced load family
<code>generators[]</code>	sheet <code>ThreePhaseGenerator</code>	missing array / sheet keeps vector empty	explicit three-phase generator family
<code>external_grids[]</code>	sheet <code>ThreePhaseExternalGrid</code>	missing array / sheet keeps vector empty	explicit sequence-based grid-equivalent family

Asynchronous motor contract: `ac.motors[]` and sheet `AsynchronousMotor`

Field	Excel column	JSON default / import rule	Meaning
<code>index</code>	<code>index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable device identifier
<code>bus</code>	<code>bus</code>	required on JSON import	connected AC bus index
<code>name</code>	<code>name</code>	default empty string	display label only
<code>in_service</code>	<code>in_service</code>	default <code>true</code>	availability flag; not a solved PF state
<code>vn_kv</code>	<code>vn_kv</code>	default 0.0	rated terminal voltage in kV
<code>sn_mva</code>	<code>sn_mva</code>	default 0.0	rated apparent-power metadata for motor fault-contribution models
<code>r_pu</code>	<code>r_pu</code>	default 0.0	motor equivalent positive-sequence resistance input
<code>x_pu</code>	<code>x_pu</code>	default 0.0	motor equivalent positive-sequence reactance input
<code>x_r</code>	<code>x_r</code>	default 0.0	X/R ratio metadata for short-circuit-style motor models
<code>lrc</code>	<code>lrc</code>	default 0.0	locked-rotor current ratio
<code>poles</code>	<code>poles</code>	default 2	machine pole count
<code>cos_phi</code>	<code>cos_phi</code>	default 0.85	rated power factor

Field	Excel column	JSON default / import rule	Meaning
efficiency	efficiency	default 0.95	rated efficiency factor
r0_pu	r0_pu	default 0.0	zero-sequence resistance input for unsymmetrical faults
x0_pu	x0_pu	default 0.0	zero-sequence reactance input for unsymmetrical faults

Three-phase bus contract: `three_phase_ac.buses[]` and sheet `ThreePhaseBus`

Field	Excel column	JSON default / import rule	Meaning
index	index	required on JSON import; blank Excel index stops table import	stable bus identifier
name	name	default empty string	bus label
bus_type	bus_type	default PQ	bus category using the same string tokens as the AC bus model
base_kv	base_kv	default 0.0	nominal line-to-line bus voltage base
in_service	in_service	default true	availability flag
vm_a_pu, vm_b_pu, vm_c_pu	same columns	default 1.0 each	per-phase initial or specified voltage magnitudes
va_a_deg, va_b_deg, va_c_deg	same columns	defaults 0.0, -120.0, 120.0	phase-angle initialization in degrees for phase order a, b, c
pd_a_mw, pd_b_mw, pd_c_mw	same columns	default 0.0	explicit per-phase active demand before projection
qd_a_mvar, qd_b_mvar, qd_c_mvar	same columns	default 0.0	explicit per-phase reactive demand before projection
gs_a_mw, gs_b_mw, gs_c_mw	same columns	default 0.0	per-phase conductance shunt terms
bs_a_mvar, bs_b_mvar, bs_c_mvar	same columns	default 0.0	per-phase susceptance shunt terms
vmin_pu	vmin_pu	default 0.9	lower per-unit voltage bound
vmax_pu	vmax_pu	default 1.1	upper per-unit voltage bound
area	area	default 1	planning / reporting area tag
zone	zone	default 1	planning / reporting zone tag

Three-phase line contract: `three_phase_ac.lines[]` and sheet `ThreePhaseLine`

Field	Excel column	JSON default / import rule	Meaning
index	index	required on JSON import; blank Excel index stops table import	stable line identifier
from_bus, to_bus	same columns	required on JSON import	line terminal bus indices

Field	Excel column	JSON default / import rule	Meaning
<code>name</code>	<code>name</code>	default empty string	line label
<code>in_service</code>	<code>in_service</code>	default <code>true</code>	availability flag
<code>length_km</code>	<code>length_km</code>	default 0.0	physical route length
<code>parallel</code>	<code>parallel</code>	default 1	number of parallel circuits represented by the row
<code>r1_ohm_per_km</code> , <code>x1_ohm_per_km</code> , <code>c1_nf_per_km</code>	same columns	default 0.0	positive-sequence per-km impedance / capacitance data
<code>r0_ohm_per_km</code> , <code>x0_ohm_per_km</code> , <code>c0_nf_per_km</code>	same columns	default 0.0	zero-sequence per-km impedance / capacitance data
<code>r1_pu</code> , <code>x1_pu</code> , <code>b1_pu</code>	same columns	default 0.0	positive-sequence per-unit branch parameters
<code>r0_pu</code> , <code>x0_pu</code> , <code>b0_pu</code>	same columns	default 0.0	zero-sequence per-unit branch parameters
<code>max_i_ka</code>	<code>max_i_ka</code>	default 0.0	thermal current limit in kA
<code>rate_a_mva</code>	<code>rate_a_mva</code>	default 0.0	apparent-power rating used by planning / screening logic
<code>failure_rate</code>	<code>failure_rate</code>	default 0.0	reliability failure-rate input
<code>mttr_hr</code>	<code>mttr_hr</code>	default 0.0	repair-time input in hours

Three-phase transformer contract: `three_phase_ac.transformers[]` and sheet `ThreePhaseTransformer`

Field	Excel column	JSON default / import rule	Meaning
<code>index</code>	<code>index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable transformer identifier
<code>hv_bus</code> , <code>lv_bus</code>	same columns	required on JSON import	high-voltage and low-voltage terminal bus indices
<code>name</code>	<code>name</code>	default empty string	transformer label
<code>in_service</code>	<code>in_service</code>	default <code>true</code>	availability flag
<code>sn_mva</code>	<code>sn_mva</code>	default 0.0	rated apparent power
<code>vn_hv_kv</code> , <code>vn_lv_kv</code>	same columns	default 0.0	rated winding voltages
<code>vk_percent</code> , <code>vkr_percent</code>	same columns	default 0.0	short-circuit magnitude and resistive share in percent
<code>pfe_kw</code> , <code>i0_percent</code>	same columns	default 0.0	no-load loss and excitation-current inputs
<code>vector_group</code>	<code>vector_group</code>	default empty string	winding connection / clock notation for zero-sequence path interpretation
<code>vk0_percent</code> , <code>vkr0_percent</code> , <code>mag0_percent</code> , <code>mag0_rx</code> , <code>si0_hv_partial</code>	same columns	defaults 0.0 except <code>si0_hv_partial=0.5</code>	zero-sequence transformer data preserved for SC and future unbalanced workflows
<code>tap_side</code> , <code>tap_pos</code> , <code>tap_min</code> , <code>tap_max</code> , <code>tap_neutral</code>	same columns	default 0	tap-changer placement and discrete tap position metadata
<code>tap_step_percent</code>	<code>tap_step_percent</code>	default 0.0	tap step size in percent
<code>shift_deg</code>	<code>shift_deg</code>	default 0.0	phase shift angle in degrees
<code>mtbf_hours</code> , <code>mttr_hours</code>	same columns	default 0.0	reliability inputs retained in the explicit contract

Three-phase load contract: `three_phase_ac.loads[]` and sheet `ThreePhaseLoad`

Field	Excel column	JSON default / import rule	Meaning
index	index	required on JSON import; blank Excel index stops table import	stable load identifier
bus	bus	required on JSON import	connected three-phase bus index
name	name	default empty string	load label
in_service	in_service	default true	availability flag
connection	connection	default "wye"	wiring model; current code expects free-form connection text such as wye or delta
grounded	grounded	default true	grounding-path assumption for the connection
p_a_mw, p_b_mw, p_c_mw	same columns	default 0.0	per-phase active demand
q_a_mvar, q_b_mvar, q_c_mvar	same columns	default 0.0	per-phase reactive demand
const_z_percent, const_i_percent	same columns	default 0.0	ZIP-model constant-impedance and constant-current shares
const_p_percent	const_p_percent	default 100.0	ZIP-model constant-power share
motor_percent	motor_percent	default 0.0	embedded motor-fraction share inside the unbalanced load object
lrc_pu	lrc_pu	default 0.0	locked-rotor current ratio for the embedded motor fraction
x_r_ratio	x_r_ratio	default 0.0	X/R ratio for the embedded motor fraction

Three-phase generator contract: `three_phase_ac.generators[]` and sheet `ThreePhaseGenerator`

Field	Excel column	JSON default / import rule	Meaning
index	index	required on JSON import; blank Excel index stops table import	stable generator identifier
bus	bus	required on JSON import	connected three-phase bus index
name	name	default empty string	generator label
in_service	in_service	default true	availability flag
is_slack	is_slack	default false	slack-machine indicator for explicit three-phase studies
p_mw, q_mvar	same columns	default 0.0	current active / reactive injection setpoints
vm_pu	vm_pu	default 1.0	terminal voltage target or initialization
pmax_mw, pmin_mw, qmax_mvar, qmin_mvar	same columns	default 0.0	capability limits
mbase_mva	mbase_mva	default 0.0	machine MVA base for the sequence reactances

Field	Excel column	JSON default / import rule	Meaning
<code>xd_pu</code> , <code>xdpp_pu</code> , <code>x2_pu</code> , <code>x0_pu</code> , <code>r0_pu</code>	same columns	default 0.0	positive / subtransient / negative / zero-sequence machine impedance set

Three-phase external-grid contract: `three_phase_ac.external_grids[]` and sheet `ThreePhaseExternalGrid`

Field	Excel column	JSON default / import rule	Meaning
<code>index</code>	<code>index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable external-grid identifier
<code>bus</code>	<code>bus</code>	required on JSON import	connected three-phase bus index
<code>name</code>	<code>name</code>	default empty string	source label
<code>in_service</code>	<code>in_service</code>	default <code>true</code>	availability flag
<code>vm_pu</code>	<code>vm_pu</code>	default 1.0	Thevenin voltage magnitude target
<code>va_deg</code>	<code>va_deg</code>	default 0.0	Thevenin voltage angle in degrees
<code>s_sc_max_mva</code> , <code>s_sc_min_mva</code>	same columns	default 0.0	short-circuit strength limits
<code>rx_max</code> , <code>rx_min</code>	same columns	default 0.0	R/X ratio bounds used in equivalent-source parametrization
<code>r1_pu</code> , <code>x1_pu</code>	same columns	default 0.0	positive-sequence Thevenin impedance
<code>r2_pu</code> , <code>x2_pu</code>	same columns	default 0.0	negative-sequence Thevenin impedance
<code>r0_pu</code> , <code>x0_pu</code>	same columns	default 0.0	zero-sequence Thevenin impedance

9.5 Compact Power-Flow Workbook Additions

Sheet updated by <code>save_results_compact.xlsx</code>	Appended columns	Current note
<code>ACBus</code>	<code>result_vm_pu</code> , <code>result_va_deg</code>	solved AC bus state.
<code>DCBus</code>	<code>result_vdc_pu</code>	solved DC voltage.
<code>ACBranch</code>	<code>result_pf_mw</code> , <code>result_qf_mvar</code> , <code>result_pt_mw</code> , <code>result_qt_mvar</code>	branch terminal powers.
<code>DCBranch</code>	<code>result_v_from_pu</code> , <code>result_v_to_pu</code> , <code>result_i_pu</code> , <code>result_p_from_mw</code> , <code>result_p_to_mw</code> , <code>result_loss_mw</code>	derived from solved DC bus voltages and branch resistance.
<code>Storage</code>	<code>result_vm_pu</code> , <code>result_va_deg</code>	bus-state annotation only; storage dispatch is not recomputed.
<code>DCStorage</code>	<code>result_vdc_pu</code>	bus-state annotation only.

Sheet updated by save_results_c ompact_xlsx	Appended columns	Current note
Load	result_vm_pu, result_va_deg	bus-state annotation on the raw AC load table.
ACLoad	result_vm_pu, result_va_deg	bus-state annotation on the aggregated AC load view.
DCLoad	result_vdc_pu	bus-state annotation on the aggregated DC load view.
ChargingStation	result_vm_pu, result_va_deg	station bus state.
VSCConverter	result_p_ac_mw, result_q_ac_mvar, result_p_dc_mw, result_loss_mw, result_v_ac_pu, result_v_dc_pu	solved converter transfer and terminal voltages.
DCDCConverter	result_p_in_mw, result_p_out_mw, result_loss_mw, result_v_in_pu, result_v_out_pu	solved converter transfer and terminal voltages.
VPP	result_vm_pu, result_va_deg	aggregation-bus annotation only.
Microgrid	result_vm_pu, result_va_deg	PCC-bus annotation only.
MobileStorage	result_vm_pu, result_va_deg	current bus annotation only.
Results_Summary	Key, Value	key-value sheet with converged, iterations, and residual.

9.6 Dedicated Result Workbooks

Dedicated power-flow result sheets

Sheet	Columns	Current note
Results_ACBus	index, name, vm_pu, va_deg	solved AC bus state.
Results_DCBus	index, name, vm_pu	this sheet uses the AC-style column name vm_pu even though the value is the DC solved voltage.
Results_Branch Flow	index, from_bus, to_bus, pf_mw, qf_mvar, pt_mw, qt_mvar	branch flows reported in branch order.
Results_DCBranch	index, name, from_bus, to_bus, in_service, v_from_pu, v_to_pu, i_pu, p_from_mw, p_to_mw, loss_mw	computed from solved voltages and branch resistance.

Sheet	Columns	Current note
Results_VSCConverter	index, name, bus_ac, bus_dc, in_service, control_mode, p_ac_mw, q_ac_mvar, p_dc_mw, loss_mw, p_set_mw, q_set_mvar, v_ac_pu, v_dc_pu, eta, loss_percent	includes both solved transfer and original setpoint metadata.
Results_DCDCConverter	index, name, bus_in, bus_out, in_service, control_mode, p_ref_mw, v_ref_pu, v_in_pu, v_out_pu, eta, p_in_mw, p_out_mw, loss_mw	includes both solved transfer and original reference metadata.
Results_Storage_AC	index, name, bus, in_service, p_mw, q_mvar, soc, soh, vm_pu, va_deg	uses soc_init as the reported soc.
Results_Storage_DC	index, name, bus, in_service, p_mw, q_mvar, soc, soh, vdc_pu	uses soc_init as the reported soc.
Results_ChargingStation	index, name, bus, in_service, num_chargers, p_total_kw, q_total_kvar, max_power_kw, utilization_rate, vm_pu, va_deg	station bus state plus current aggregate demand fields.
Results_Charger	index, name, station_id, station_bus, in_service, charger_type, p_rated_kw, p_ch_max_kw, p_dis_max_kw, eta, v2g_capable, vm_pu, va_deg	charger_type now uses the same string token family as JSON and workbook input sheets.
Results_EnergyRouter	index, name, in_service, router_type, num_ports, p_rated_mw, loss_percent, p_port_sum_mw, q_port_sum_mvar	device-level summary only; solved per-port values live in the port sheet.
Results_EnergyRouterPort	router_index, port_index, port_name, bus, port_type, control_mode, in_service, p_mw, q_mvar, v_solved_pu	flattens nested router ports into one table.
Results_VPP	index, name, aggregation_bus, in_service, p_output_mw, q_output_mvar, pmax_mw, pmin_mw, vm_pu, va_deg	aggregation-bus annotation only.
Results_Microgrid	index, name, pcc_bus, operating_mode, in_service, p_exchange_mw, total_generation_mw, total_load_mw, vm_pu, va_deg, internal_buses	internal_buses is written as one comma-separated string.
Results_MobileStorage	index, name, bus, status, in_service, p_mw, q_mvar, soc, target_bus, vm_pu, va_deg	uses soc_init as the reported soc.
Results_Summary	Key, Value	keys are converged, iterations, and residual.

Result-Sheet Enum and Flag Conventions

Artifact	Field	Output representation	Field note
Results_VSCConverter	control_mode	string token: PQ, VDC_Q, VDC_VAC	copied from the input converter mode; not a solver-selected mode switch
Results_DCDCConverter	control_mode	string token: Voltage, Power, Droop	copied from the input converter mode
Results_Charger	charger_type	string token: AC_L1, AC_L2, DC_Fast	now aligned with JSON and workbook input instead of numeric enum codes
Results_EnergyRouterPort	port_type	string token: AC or DC	physical port side classification
Results_EnergyRouterPort	control_mode	string token: PQ, VF, Droop	copied from the input port control mode
Results_Microgrid	operating_mode	string token: Grid Connected, Islanded, Transition	reported operating mode metadata, not a PF state variable
Results_MobileStorage	status	string token: Stationary, InTransit, Deployed	reported mobility state metadata
Results_Charger	v2g_capable	boolean token: TRUE / FALSE	static capability flag, not current dispatch direction
Results_* device sheets	in_service	boolean token: TRUE / FALSE	copied device availability flag across result workbooks
Results_Summary, OPF_Summary	converged	boolean token: TRUE / FALSE	solver convergence outcome

Dedicated OPF and carbon result sheets

Sheet	Columns	Current note
OPF_Bus	index, name, base_kv, vm_pu, va_deg, dpd_mw, dqd_mvar	bus voltages plus bus-level load shedding vectors.
OPF_Generator	index, name, bus, pg_mw, qg_mvar, pmax_mw, pmin_mw, qmax_mvar, qmin_mvar, cost_c2, cost_c1, cost_c0, fuel_type	generator dispatch report.
OPF_Converter	index, name, bus_ac, bus_dc, pac_mw, qac_mvar	only VSC converters are reported here.

Sheet	Columns	Current note
OPF_Summary	Key, Value	keys are converged, iterations, outer_iterations, objective, max_constraint_violation, max_stationarity, and status.
LoadCarbon	load_index, bus, demand_mw, carbon_intensity_tco2_mwh, total_emissions_tco2	dedicated carbon workbook, sheet 1.
DCLoadCarbon	load_index, bus, demand_mw, carbon_intensity_tco2_mwh, total_emissions_tco2	dedicated carbon workbook, sheet 2.
BusCarbon	bus_index, carbon_intensity_tco2_mwh	dedicated carbon workbook, sheet 3.
DCBusCarbon	bus_index, carbon_intensity_tco2_mwh	dedicated carbon workbook, sheet 4.
BranchCarbon	branch_index, from_bus, to_bus, loss_mw, carbon_intensity_tco2_mwh, total_emissions_tco2	dedicated carbon workbook, sheet 5.
DCBranchCarbon	branch_index, from_bus, to_bus, loss_mw, carbon_intensity_tco2_mwh, total_emissions_tco2	dedicated carbon workbook, sheet 6.
VSCCarbon	converter_index, bus_ac, bus_dc, ac_to_dc, input_power_mw, output_power_mw, loss_mw, carbon_intensity_tco2_mwh, total_emissions_tco2	dedicated carbon workbook, sheet 7.
DCDCCarbon	converter_index, bus_in, bus_out, input_to_output, input_power_mw, output_power_mw, loss_mw, carbon_intensity_tco2_mwh, total_emissions_tco2	dedicated carbon workbook, sheet 8.
StorageCarbon	storage_index, bus, is_dc, p_mw, soc, stored_energy_mwh, soc_carbon_intensity_tco2_mwh, carbon_intensity_tco2_mwh, total_emissions_tco2	dedicated carbon workbook sheet for explicit storage charge/discharge carbon semantics.
EnergyRouterCarbon	router_index, input_power_mw, output_power_mw, loss_mw, active_input_ports, active_output_ports, carbon_intensity_tco2_mwh, total_emissions_tco2	dedicated carbon workbook sheet for router-internal loss allocation reconstructed from port transfers.

Sheet	Columns	Current note
Summary	key-value blocks for tracing and matrix summaries	dedicated carbon workbook summary sheet; contains generation/load/loss totals, balance errors, and flags such as <code>tracing_verified</code> and <code>matrix_solved</code> .

9.7 Current I/O Coverage Gaps That Matter to Engineering Work

Public model area	JSON system file	Excel input workbook	Engineering consequence
ACBus, ACBranch, DCBus, DCBranch	yes	yes	core network objects are available in both I/O families.
Generator, Load, Shunt, ExternalGrid	yes	yes	Excel coverage is narrower than JSON for the richer machine / source metadata.
StaticGenerator, RenewableGen, PVSystem, Storage	yes	yes	Excel exposes only the operational subset; JSON is the better notebook-to-code contract for detailed studies.
Transformer2W	yes	yes	Excel omits zero-sequence and reliability fields.
Transformer3W	yes	yes	JSON and Excel now both carry the explicit 3-winding transformer data, although solver use still goes through projection.
FlexibleLoad, AsymmetricLoad	yes	yes	both I/O families now preserve the rich pre-projection load models.
ChargingStation	yes	yes	input support exists in both families.
Charger	yes	yes	both I/O families now preserve per-charger metadata; the parent station link remains <code>station_id</code> .
DCLoad	yes	yes	the main system JSON and Excel workbook are now aligned for the explicit DC load table.
VSCConverter, DCDCCConverter	yes	yes	Excel coverage is narrower than JSON.
EnergyRouter, EnergyRouterPort	yes	yes	router studies can now use either JSON or Excel input, with ports flattened into a dedicated Excel table.
VPP, Microgrid, MobileStorage	yes	yes	Excel again keeps only a reduced operational subset.
AsynchronousMotor	yes	yes	the SC motor model now has a stable external contract in both families.

Public model area	JSON system file	Excel input workbook	Engineering consequence
ThreePhaseACSystem family	yes	yes	the optional unbalanced subsystem can now round-trip through both JSON and Excel, although main PF/OPF/SC still consume it after projection.

9.8 Interpretation

The current I/O layer is already useful, but it is not a single canonical schema yet.

1. JSON is now the richest external representation for the full current `HybridPowerSystem` container, including `ac.motors` and optional `three_phase_ac`.
2. Excel is an engineering workbook interface with deliberate aggregation and reduced field coverage.
3. The JSON and Excel paths are now aligned for `DCLoad`, `Transformer3W`, `FlexibleLoad`, `AsymmetricLoad`, `Charger`, `EnergyRouter`, `AsynchronousMotor`, and the `ThreePhaseACSystem` family.
4. The remaining gaps are now field-level rather than family-level, especially the SC-specific motor-fraction fields embedded inside `Load` and the lack of dedicated result-workbook contracts for explicit three-phase solved states.

10 Detailed I/O Field Dictionaries

This section expands the per-field contract to the remaining public model families. Section 02a_io_contracts already gives detailed field maps for `AsynchronousMotor`, the optional `three_phase_ac` family, and the result-sheet enum conventions. The tables below complete the same JSON-key / Excel-column dictionary pattern for the other current input models.

When the “Excel sheet / column” entry says `none`, the engineering workbook does not preserve that JSON field directly. When two workbook sheet names are listed, the first is the preferred explicit input table and the second is a fallback or derived view with the same column family.

10.1 System Metadata

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>name</code>	<code>Info[name]</code>	default "HybridAC/DC System"	top-level <code>HybridPowerSystem</code> display name
<code>base_mva</code>	<code>Info[base_mva]</code>	default 100.0	top-level system power base; workbook import also copies it into <code>ac.base_mva</code> and <code>dc.base_mva</code>
<code>ac.name</code>	<code>none</code>	default "ACSystem"	JSON-only AC subsystem label

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>ac.base_mva</code>	<code>none</code>	default 100.0	JSON-only AC subsystem base; workbook uses the shared top-level <code>base_mva</code> instead
<code>ac.freq_hz</code>	<code>Info[ac_freq_hz]</code>	default 50.0	nominal AC frequency in Hz
<code>dc.name</code>	<code>none</code>	default "DCSystem"	JSON-only DC subsystem label
<code>dc.base_mva</code>	<code>none</code>	default 100.0	JSON-only DC subsystem base; workbook again uses the shared top-level <code>base_mva</code>
<code>three_phase_ac.*</code>	<code>Info[three_phase_*]</code> plus dedicated sheets	see Section 02a_io_contracts	optional explicit unbalanced subsystem documented in the earlier detailed dictionary

10.2 AC Network and Demand Models

ACBus : `ac.buses[]` / sheet **ACBus**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code>	<code>ACBus.index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable bus identifier
<code>name</code>	<code>ACBus.name</code>	default empty string	bus label
<code>bus_type</code>	<code>ACBus.bus_type</code>	default PQ	bus category token PQ/PV/SLACK/ISOLATED
<code>pd_mw, qd_mvar</code>	same columns	default 0.0	embedded bus demand snapshot; workbook export may contain synchronized aggregated demand rather than original raw struct values
<code>vm_pu, va_deg</code>	same columns	defaults 1.0, 0.0	initial or specified bus voltage magnitude and angle
<code>base_kv</code>	<code>ACBus.base_kv</code>	default 110.0	nominal bus voltage base
<code>vmin_pu, vmax_pu</code>	same columns	defaults 0.9, 1.1	voltage operating bounds
<code>gs_mw, bs_mvar</code>	same columns	default 0.0	shunt conductance and susceptance embedded at the bus
<code>area, zone</code>	same columns	defaults 1, 1	planning / reporting partitions
<code>in_service</code>	<code>ACBus.in_service</code>	default true	availability flag
<code>n_customers</code>	<code>ACBus.n_customers</code>	default 0	customer-count metadata
<code>importance</code>	<code>ACBus.importance</code>	default 1.0	bus criticality / weighting metadata
<code>latitude, longitude</code>	same columns	default 0.0	geospatial metadata only

ACBranch : `ac.branches[]` / sheet **ACBranch**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code>	<code>ACBranch.index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable branch identifier
<code>from_bus, to_bus</code>	same columns	required on JSON import	branch terminal bus indices
<code>name</code>	<code>ACBranch.name</code>	default empty string	branch label
<code>r_pu, x_pu, b_pu</code>	same columns	default 0.0	series resistance, series reactance, and total shunt charging in per unit
<code>tap</code>	<code>ACBranch.tap</code>	default 1.0	off-nominal tap ratio used in the branch model
<code>shift_deg</code>	<code>ACBranch.shift_deg</code>	default 0.0	phase-shifting angle in degrees
<code>rate_a_mva, rate_b_mva, rate_c_mva</code>	same columns	default 0.0	thermal / planning ratings
<code>length_km</code>	<code>ACBranch.length_km</code>	default 0.0	physical route length metadata
<code>in_service</code>	<code>ACBranch.in_service</code>	default <code>true</code>	availability flag
<code>failure_rate, mttr_hr</code>	same columns	default 0.0	reliability metadata
<code>r0_pu, x0_pu, b0_pu, vn_hv_kv, vn_lv_kv, sn_mva</code>	<code>none</code>	not part of the current JSON / Excel contract	struct-level short-circuit and transformer-nameplate helpers currently stay internal to the in-memory model

Load : `ac.loads[]` / sheet Load (fallback `ACLoad`)

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code>	<code>Load.index</code> or <code>ACLoad.index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable load identifier
<code>bus</code>	<code>Load.bus</code> or <code>ACLoad.bus</code>	required on JSON import	connected AC bus index
<code>name</code>	same column	default empty string	load label
<code>in_service</code>	same column	default <code>true</code>	availability flag
<code>p_mw, q_mvar</code>	same columns	default 0.0	active and reactive demand
<code>model</code>	same column	default <code>ConstantPower</code>	ZIP / exponential / constant-power load-model token
<code>z_percent_p, i_percent_p, p_percent_p</code>	same columns	defaults 0.0, 0.0, 100.0	active-power ZIP shares
<code>z_percent_q, i_percent_q, p_percent_q</code>	same columns	defaults 0.0, 0.0, 100.0	reactive-power ZIP shares
<code>controllable</code>	same column	default <code>false</code>	whether shedding / dispatch logic may treat the load as controllable
<code>p_min_mw</code>	<code>none</code>	default 0.0	JSON-only minimum retained active demand; workbook currently omits this field for AC loads
<code>cost_mw</code>	same column	default 0.0	marginal shedding or curtailment penalty

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
priority	same column	default Medium	service-priority token Low/Medium/High/Critical
n_customers	none	default 0	JSON-only customer-count metadata
profile_id	none	default -1	JSON-only profile linkage identifier
sn_mva, motor_percent, x_sub_pu, r_sc_pu	none	not part of the current main JSON contract	short-circuit helper fields remain outside the stable Load I/O contract

FlexibleLoad : ac.flexible_loads[] / sheet FlexibleLoad

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected AC bus
name	FlexibleLoad.name	default empty string	load label
in_service	FlexibleLoad.in_service	default true	availability flag
p_mw, q_mvar	same columns	default 0.0	baseline active and reactive demand
flex_up_mw, flex_down_mw	same columns	default 0.0	upward / downward flexibility envelope in MW
flex_duration_h	same column	default 0.0	duration for which the flex envelope can be sustained
response_time_s	same column	default 0.0	response latency in seconds
ramp_rate_mw_min	same column	default 0.0	adjustment ramp-rate limit
availability_pct	same column	default 100.0	expected availability of the flexibility service
controllable	same column	default true	whether the flexibility service is enabled
priority	same column	default Medium	load-service priority token
control_area	same column	default empty string	operator / aggregation area tag

AsymmetricLoad : ac.asymmetric_loads[] / sheet AsymmetricLoad

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected AC bus
name	AsymmetricLoad.name	default empty string	load label
in_service	AsymmetricLoad.in_service	default true	availability flag
connection	AsymmetricLoad.connection	default empty string	wiring model string such as wye or delta
grounded	AsymmetricLoad.grounded	default true	grounding-path assumption
pa_rated_mw, pb_rated_mw, pc_rated_mw	same columns	default 0.0	rated per-phase active demand

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
qa_rated_mvar, qb_rated_mvar, qc_rated_mvar	same columns	default 0.0	rated per-phase reactive demand
pa_mw, pb_mw, pc_mw	same columns	default 0.0	current per-phase active demand
qa_mvar, qb_mvar, qc_mvar	same columns	default 0.0	current per-phase reactive demand
scaling	AsymmetricLoad.scaling	default 1.0	overall scaling factor applied to the phase values
const_z_percent, const_i_percent	same columns	default 0.0	ZIP constant-impedance and constant-current shares
const_p_percent	same column	default 100.0	ZIP constant-power share
controllable	same column	default false	controllability flag
priority	same column	default Medium	load-service priority token

Shunt : `ac.shunts[]` / sheet **Shunt**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected AC bus
name	Shunt.name	default empty string	shunt label
in_service	Shunt.in_service	default true	availability flag
gs_mw, bs_mvar	same columns	default 0.0	fixed shunt conductance and susceptance
switchable	Shunt.switchable	default false	whether the shunt is modeled as stepped / switchable
n_steps	Shunt.n_steps	default 1	number of available discrete shunt steps
current_step	Shunt.current_step	default 1	currently selected shunt step
bs_per_step	Shunt.bs_per_step	default 0.0	susceptance increment per step

ExternalGrid : `ac.external_grids[]` / sheet **ExternalGrid**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected AC bus
name	ExternalGrid.name	default empty string	source label
in_service	ExternalGrid.in_service	default true	availability flag
vm_pu, va_deg	same columns	defaults 1.0, 0.0	voltage magnitude and angle target for the slack / Thevenin source
s_sc_max_mva, s_sc_min_mva	same columns	default 0.0	short-circuit strength range

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>rx_max, rx_min</code>	same columns	default 0.0	R/X ratio bounds for equivalent-source construction
<code>r_pu, x_pu</code>	none	default 0.0	JSON-only explicit positive-sequence Thevenin impedance
<code>r0_pu, x0_pu</code>	none	default 0.0	JSON-only zero-sequence source impedance
<code>vn_kv</code>	none	default 0.0	JSON-only nominal source voltage rating
<code>ikq_ka, x_r</code>	none	not part of the current JSON / Excel contract	struct-level IEC 60909 helper fields currently stay internal and are not preserved by either import path
<code>controllable</code>	<code>ExternalGrid.controllable</code>	default true	whether optimization logic may vary the source dispatch

Transformer2W : `ac.transformers_2w[]` / sheet Transformer2W

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index, hv_bus, lv_bus</code>	corresponding columns	required on JSON import; blank Excel <code>index</code> stops table import	stable identifier and terminal bus indices
<code>name</code>	<code>Transformer2W.name</code>	default empty string	transformer label
<code>std_type</code>	none	default empty string	JSON-only library / standard type name
<code>in_service</code>	<code>Transformer2W.in_service</code>	default true	availability flag
<code>sn_mva</code>	<code>Transformer2W.sn_mva</code>	default 0.0	rated apparent power
<code>vn_hv_kv, vn_lv_kv</code>	same columns	default 0.0	winding rated voltages
<code>vk_percent, vkr_percent</code>	same columns	default 0.0	short-circuit magnitude and resistive share in percent
<code>pk_kw</code>	none	default 0.0	JSON-only load-loss field retained in the richer transformer contract
<code>pfe_kw, i0_percent</code>	same columns	default 0.0	no-load loss and excitation current
<code>tap_side, tap_pos, tap_min, tap_max, tap_neutral</code>	same columns	default 0	tap-changer placement and discrete tap metadata
<code>tap_step_percent</code>	<code>Transformer2W.tap_step_percent</code>	default 0.0	tap step size in percent
<code>shift_deg</code>	<code>Transformer2W.shift_deg</code>	default 0.0	phase-shift angle in degrees
<code>vector_group</code>	<code>Transformer2W.vector_group</code>	default empty string	winding connection / clock notation
<code>z0_percent, x0_r0</code>	none	default 0.0	JSON-only zero-sequence impedance parameters
<code>mtbf_hours, mtrt_hours</code>	none	default 0.0	JSON-only reliability metadata

Transformer3W : ac.transformers_3w[] / sheet Transformer3W

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, hv_bus, mv_bus, lv_bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and three winding bus indices
name	Transformer3W.name	default empty string	transformer label
std_type	Transformer3W.std_type	default empty string	library / standard type name
in_service	Transformer3W.in_service	default true	availability flag
sn_hv_mva, sn_mv_mva, sn_lv_mva	same columns	default 0.0	winding ratings on the HV / MV / LV sides
vn_hv_kv, vn_mv_kv, vn_lv_kv	same columns	default 0.0	winding rated voltages
vk_hv_mv_percent, vk_hv_lv_percent, vk_mv_lv_percent	same columns	default 0.0	pairwise short-circuit magnitudes
vkr_hv_mv_percent, vkr_hv_lv_percent, vkr_mv_lv_percent	same columns	default 0.0	pairwise resistive shares
pfe_kw, i0_percent	same columns	default 0.0	no-load loss and excitation-current data
tap_side, tap_pos	same columns	default 0	tap-changer side and current position
tap_step_percent	Transformer3W.tap_step_percent	default 0.0	tap step size in percent
shift_mv_deg, shift_lv_deg	same columns	default 0.0	phase shifts from HV to MV / LV windings
mtbf_hours, mtr_hours	same columns	default 0.0	reliability metadata preserved in both JSON and workbook

Switch : ac.switches[] / sheet Switch

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus_from, bus_to	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and terminal bus indices
name	Switch.name	default empty string	switch label
in_service	Switch.in_service	default true	availability flag
switch_type	Switch.switch_type	default CircuitBreaker	switching-device token such as Recloser or Fuse
closed	Switch.closed	default true	topology status flag
r_contact_ohm, z_ohm	none	default 0.0	JSON-only detailed contact / arc impedance inputs
i_rated_ka, i_breaking_ka	none	default 0.0	JSON-only current capability ratings
element_type, element_id	none	defaults 0, 0	JSON-only foreign-key reference to the protected element

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>is_remote</code> , <code>is_automated</code>	none	default false	JSON-only automation metadata
<code>t_operation_s</code> , <code>p_sw_fail</code>	none	default 0.0	JSON-only operation timing and failure-probability metadata
<code>mtbf_hours</code> , <code>mttr_hours</code>	none	default 0.0	JSON-only reliability metadata

CircuitBreaker : `ac.circuit_breakers[]` / sheet `CircuitBreaker`

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code> , <code>bus_from</code> , <code>bus_to</code>	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and terminal bus indices
<code>name</code>	<code>CircuitBreaker.name</code>	default empty string	breaker label
<code>in_service</code>	<code>CircuitBreaker.in_service</code>	default true	availability flag
<code>breaker_type</code>	<code>CircuitBreaker.breaker_type</code>	default CB	breaker-type token parsed by breaker_type_from_str
<code>closed</code>	<code>CircuitBreaker.closed</code>	default true	topology status flag
<code>z_ohm</code>	<code>CircuitBreaker.z_ohm</code>	default 0.0	AC breaker impedance
<code>rated_voltage_kv</code> , <code>i_rated_ka</code> , <code>i_breaking_ka</code>	same columns	default 0.0	voltage and current ratings
<code>element_type</code> , <code>element_id</code>	same columns	defaults empty string, 0	optional protected-element reference

ChargingStation : `ac.charging_stations[]` / sheet `ChargingStation`

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code> , <code>bus</code>	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected AC bus
<code>name</code>	<code>ChargingStation.name</code>	default empty string	station label
<code>location</code>	none	default empty string	JSON-only free-form location descriptor
<code>in_service</code>	<code>ChargingStation.in_service</code>	default true	availability flag
<code>n_fast</code> , <code>n_slow</code> , <code>num_chargers</code>	same columns	default 0	counts of charger assets by class and in total
<code>p_fast_max_kw</code> , <code>p_slow_max_kw</code>	same columns	default 0.0	per-charger maximum power by service class
<code>max_power_kw</code>	same column	default 0.0	aggregate station power ceiling
<code>simultaneity_factor</code>	same column	default 1.0	diversification factor used in the aggregate station model
<code>power_factor</code>	same column	default 0.95	station-level reactive-power assumption

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>utilization_rate</code>	<code>none</code>	default 0.0	JSON-only expected utilization metadata
<code>p_total_kw, q_total_kvar</code>	same columns	default 0.0	aggregate present demand of the station
<code>mtbf_hours, mttr_hours</code>	<code>none</code>	default 0.0	JSON-only reliability metadata

Charger : `ac.chargers[]` / sheet **Charger**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code>	<code>Charger.index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable charger identifier
<code>name</code>	<code>Charger.name</code>	default empty string	charger label
<code>station_id</code>	<code>Charger.station_id</code>	default 0	parent charging-station foreign key
<code>charger_type</code>	<code>Charger.charger_type</code>	default <code>AC_L2</code>	charger-class token <code>AC_L1/AC_L2/DC_Fast</code>
<code>in_service</code>	<code>Charger.in_service</code>	default <code>true</code>	availability flag
<code>pRated_kw</code>	<code>Charger.pRated_kw</code>	default 0.0	rated charger power
<code>pCh_max_kw, pCh_min_kw</code>	same columns	default 0.0	charging power limits
<code>eta</code>	<code>Charger.eta</code>	default 0.95	charger conversion efficiency
<code>v2g_capable</code>	<code>Charger.v2g_capable</code>	default <code>false</code>	whether reverse power flow is allowed
<code>pDis_max_kw</code>	<code>Charger.pDis_max_kw</code>	default 0.0	maximum discharging power when V2G is enabled
<code>mtbf_hours, mttr_hours</code>	same columns	default 0.0	reliability metadata preserved in both JSON and workbook

10.3 Generation and Storage Models

Generator : `ac.generators[]` / sheet **Generator**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index, bus</code>	corresponding columns	required on JSON import; blank Excel <code>index</code> stops table import	stable identifier and connected AC bus
<code>name</code>	<code>Generator.name</code>	default empty string	generator label
<code>in_service</code>	<code>Generator.in_service</code>	default <code>true</code>	availability flag
<code>pg_mw, qg_mvar</code>	same columns	default 0.0	active and reactive generation setpoints
<code>vg_pu</code>	<code>Generator.vg_pu</code>	default 1.0	voltage setpoint on the connected bus
<code>pmax_mw, pmin_mw, qmax_mvar, qmin_mvar</code>	same columns	default 0.0	operating capability limits
<code>is_slack</code>	<code>Generator.is_slack</code>	default <code>false</code>	slack-machine flag

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>fuel_type</code>	<code>Generator.fuel_type</code>	default <code>Unknown</code>	fuel-type token such as <code>Gas</code> , <code>Wind</code> , or <code>Solar</code>
<code>cost_c2</code> , <code>cost_c1</code> , <code>cost_c0</code>	same columns	default 0.0	quadratic production-cost coefficients
<code>startup_cost</code> , <code>shutdown_cost</code> , <code>min_up_time_hr</code> , <code>min_dn_time_hr</code>	<code>none</code>	default 0.0	JSON-only unit-commitment metadata
<code>ramp_up_mw_min</code> , <code>ramp_dn_mw_min</code>	same columns	default 0.0	ramp-rate limits
<code>mbase_mva</code> , <code>inertia_h</code> , <code>droop_r</code>	same columns	defaults 0.0, 0.0, 0.05	machine base, inertia, and governor droop metadata
<code>xd_pu</code> , <code>xdp_pu</code> , <code>xdpp_pu</code> , <code>td0p_s</code> , <code>td0pp_s</code>	<code>none</code>	default 0.0	JSON-only machine dynamic / short-circuit parameters
<code>ra_pu</code> , <code>vn_kv</code> , <code>cos_phi</code> , <code>xq_pu</code> , <code>xd_xq</code> , <code>x0_pu</code> , <code>r0_pu</code>	<code>none</code>	not part of the current JSON / Excel contract	additional short-circuit parameters exist in the C++ struct but are not yet serialized or workbook-backed
<code>emission_factor_tco2_mwh</code> , <code>nox_factor_kg_mwh</code> , <code>so2_factor_kg_mwh</code>	<code>none</code>	default 0.0	JSON-only emissions metadata
<code>forced_outage_rate</code> , <code>mtr_hr</code>	<code>none</code>	default 0.0	JSON-only reliability metadata

StaticGenerator : `ac.static_generators[]` or `dc.static_generators[]` / sheets
StaticGenerator and **DCStaticGen**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code> , <code>bus</code>	corresponding columns	required on JSON import; blank Excel <code>index</code> stops table import	stable identifier and connected AC or DC bus
<code>name</code>	corresponding <code>name</code> column	default empty string	generator label
<code>in_service</code>	corresponding <code>in_service</code> column	default <code>true</code>	availability flag
<code>sgen_type</code>	corresponding <code>sgen_type</code> column	default <code>Other</code>	generator-type token such as <code>Wind</code> , <code>Battery</code> , or <code>FuelCell</code>
<code>p_mw</code> , <code>q_mvar</code>	same columns	default 0.0	present active / reactive injection
<code>p_rated_mw</code> , <code>sn_mva</code>	same columns	default 0.0	rated active power and apparent power
<code>pmax_mw</code> , <code>pmin_mw</code> , <code>qmax_mvar</code> , <code>qmin_mvar</code>	same columns	default 0.0	operating capability limits
<code>scaling</code>	same column	default 1.0	scalar multiplier for the static generation profile
<code>controllable</code>	same column	default <code>false</code>	controllability flag
<code>v_ref_pu</code> , <code>k_p</code> , <code>k_q</code> , <code>k_rx</code>	<code>none</code>	defaults 1.0, 0.0, 0.0, 1.0, 0.0	JSON-only control and equivalent-source tuning parameters
<code>co2_emission_rate</code>	<code>none</code>	default 0.0	JSON-only emissions metadata

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
mtbf_hours, mttr_hours	none	default 0.0	JSON-only reliability metadata

Storage : `ac.storage[]` or `dc.storage[]` / sheets **Storage** and **DCStorage**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected bus
name	corresponding name column	default empty string	storage label
in_service	corresponding in_service column	default true	availability flag
p_mw, q_mvar	same columns	default 0.0	present active and reactive dispatch
p_rated_mw	same column	default 0.0	rated converter power
pmax_mw, pmin_mw, qmax_mvar, qmin_mvar	same columns	default 0.0	operating limits
e_rated_mwh	same column	default 0.0	energy capacity
soc_init, soc_min, soc_max	same columns	defaults 0.5, 0.1, 0.9	initial and admissible state-of-charge values
soc_carbon_intensity_tco2_mwh	same column	default 0.0	explicit carbon intensity of the stored energy state used by carbon-flow analysis during discharge
eta_charge, eta_discharge	same columns	default 0.95	charging and discharging efficiency
self_discharge_pct	none	default 0.0	JSON-only storage leakage metadata
max_cycles, current_cycles	none	defaults 5000, 0	JSON-only lifecycle counters
soh	none	default 1.0	JSON-only state-of-health value
replacement_cost	none	default 0.0	JSON-only replacement-cost metadata
forced_outage_rate, mttr_hr	none	default 0.0	JSON-only reliability metadata

RenewableGen : `ac.renewable_gens[]` / sheet **RenewableGen**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected AC bus
name	RenewableGen.name	default empty string	generator label
in_service	RenewableGen.in_service	default true	availability flag
type	RenewableGen.type	default Wind	renewable-type token such as Wind , SolarPV , or Hydro

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
p_mw, q_mvar	same columns	default 0.0	present active and reactive output
p_rated_mw	same column	default 0.0	rated active power
qmax_mvar, qmin_mvar	same columns	default 0.0	reactive capability limits
curtailable	same column	default true	whether curtailment is allowed
cost_curtail_mwh	none	default 0.0	JSON-only curtailment penalty
capacity_factor	RenewableGen.capacity_factor	default 0.3	expected normalized availability
profile_id	none	default -1	JSON-only profile linkage identifier
emission_offset_tco2_mwh	none	default 0.0	JSON-only avoided-emissions metadata

PVSystem : ac.pv_systems[] / sheet PVSystem

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected AC bus
name	PVSystem.name	default empty string	PV system label
in_service	PVSystem.in_service	default true	availability flag
p_mw, q_mvar	same columns	default 0.0	present active and reactive output
sn_mva	PVSystem.sn_mva	default 0.0	apparent-power rating
pmax_mw, pmin_mw, qmax_mvar, qmin_mvar	same columns	default 0.0	operating capability limits
control_mode	PVSystem.control_mode	default MPPT	inverter control-mode token; one of MPPT, VoltVar, VoltWatt, FixedPF, or Curtailed
controllable	none	default false	JSON-only controllability flag
v_ac_set_pu, v_dc_set_pu	none	default 1.0	JSON-only AC/DC voltage reference targets
inverter_eff	PVSystem.inverter_eff	default 0.97	inverter efficiency
loss_percent	none	default 0.0	JSON-only miscellaneous loss factor
num_series, num_parallel, vmpp, impp, voc, isc, alpha_isc, beta_voc	none	default 0 or 0.0	JSON-only panel-array electrical metadata
irradiance, temperature	none	defaults 1000.0, 25.0	JSON-only environmental inputs
mtbf_panel_hours, mttr_panel_hours, mtbf_inverter_hours, mttr_inverter_hours	none	default 0.0	JSON-only reliability metadata

10.4 DC and Hybrid Coordination Models

DCBus : `dc.buses[]` / sheet **DCBus**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code>	<code>DCBus.index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable DC bus identifier
<code>name</code>	<code>DCBus.name</code>	default empty string	bus label
<code>bus_type</code>	<code>DCBus.bus_type</code>	default <code>DC_P</code>	DC bus-type token <code>DC_P</code> or <code>DC_V</code>
<code>vm_pu</code>	<code>DCBus.vm_pu</code>	default 1.0	DC voltage magnitude in per unit
<code>vmax_pu</code> , <code>vmin_pu</code>	same columns	defaults 1.1, 0.9	DC voltage operating bounds
<code>pd_mw</code>	<code>DCBus.pd_mw</code>	default 0.0	embedded DC demand snapshot; workbook export may reflect synchronized aggregate demand
<code>in_service</code>	<code>DCBus.in_service</code>	default <code>true</code>	availability flag
<code>latitude</code> , <code>longitude</code>	<code>none</code>	defaults 0.0; import also accepts <code>lat/lon</code> aliases	JSON-only coordinates
<code>base_kv</code> , <code>area</code> , <code>zone</code> , <code>n_customers</code> , <code>importance</code> , <code>is_load</code>	<code>none</code>	not part of current JSON / Excel contract	C++ struct fields that are not preserved by either serializer

DCBranch : `dc.branches[]` / sheet **DCBranch**

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code> , <code>from_bus</code> , <code>to_bus</code>	corresponding columns	required on JSON import; blank Excel <code>index</code> stops table import	stable identifier and terminal bus indices
<code>name</code>	<code>DCBranch.name</code>	default empty string	branch label
<code>r_pu</code>	<code>DCBranch.r_pu</code>	default 0.0	branch resistance in per unit
<code>rate_a_mva</code>	<code>DCBranch.rate_a_mva</code>	default 0.0	branch rating
<code>length_km</code>	<code>DCBranch.length_km</code>	default 0.0	physical route length metadata
<code>in_service</code>	<code>DCBranch.in_service</code>	default <code>true</code>	availability flag
<code>base_kv</code> , <code>s_max_mva</code> , <code>n_parallel</code> , <code>mtbf_hours</code> , <code>mttr_hours</code> , <code>t_scheduled_hr</code>	<code>none</code>	not part of current JSON / Excel contract	C++ struct fields that are not preserved by either serializer

DCLoad : `dc.loads[]` / sheet **DCLoadRaw** (fallback **DCLoad**)

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected DC bus
name	same column	default empty string	load label
in_service	same column	default true	availability flag
p_mw	same column	default 0.0	active demand
controllable	same column	default false	controllability flag
p_min_mw	same column	default 0.0	minimum retained demand under curtailment
cost_mw	same column	default 0.0	marginal curtailment penalty
profile_id	none	default -1	JSON-only profile linkage identifier

StaticGeneratorDC : dc.dc_static_generators[] / sheet DCStaticGenerator

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected DC bus
name	DCStaticGenerator.name	default empty string	source label
in_service	DCStaticGenerator.in_service	default true	availability flag
type	DCStaticGenerator.type	default empty string	free-form DC source type
p_set_mw	DCStaticGenerator.p_set_mw	default 0.0	active-power setpoint
scaling	DCStaticGenerator.scaling	default 1.0	profile scaling factor
profile_id	DCStaticGenerator.profile_id	default -1	profile linkage identifier
pmax_mw, pmin_mw	same columns	default 0.0	operating limits
controllable	DCStaticGenerator.controllable	default false	controllability flag
mtbf_hours, mttr_hours	Excel columns mtbf_hr , mttr_hr	default 0.0	reliability metadata; Excel uses the hr suffix
t_scheduled_hr	same column	default 0.0	scheduled outage duration metadata

PVArrayDC : dc.pv_arrays[] / sheet DCPVArray

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
index, bus	corresponding columns	required on JSON import; blank Excel index stops table import	stable identifier and connected DC bus
name	DCPVArray.name	default empty string	DC PV array label
in_service	DCPVArray.in_service	default true	availability flag
p_set_mw	DCPVArray.p_set_mw	default 0.0	active-power setpoint

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>profile_id</code>	<code>DCPVArray.profile_id</code>	default -1	profile linkage identifier
<code>num_series, num_parallel</code>	same columns	default 0	array topology
<code>vmpp, impp, voc, isc</code>	same columns	default 0.0	module electrical parameters
<code>alpha_isc, beta_voc</code>	same columns	default 0.0	temperature coefficients
<code>temperature, irradiance</code>	same columns	defaults 25.0, 1000.0	operating condition inputs
<code>mtbf_hours, mttr_hours</code>	Excel columns <code>mtbf_hr</code> , <code>mttr_hr</code>	default 0.0	reliability metadata; Excel uses the <code>hr</code> suffix
<code>t_scheduled_hr</code>	same column	default 0.0	scheduled outage duration metadata

VSCConverter : `vsc_converters[]` / sheet VSCConverter

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index, bus_ac, bus_dc</code>	corresponding columns	required on JSON import; blank Excel <code>index</code> stops table import	stable identifier and AC/DC terminal bus indices
<code>name</code>	<code>VSCConverter.name</code>	default empty string	converter label
<code>in_service</code>	<code>VSCConverter.in_service</code>	default <code>true</code>	availability flag
<code>control_mode</code>	<code>VSCConverter.control_mode</code>	default PQ	converter-mode token PQ/VD C_Q/VDC_VAC
<code>p_set_mw, q_set_mvar</code>	same columns	default 0.0	AC-side active and reactive setpoints
<code>v_dc_set_pu, v_ac_set_pu</code>	same columns	default 1.0	DC and AC voltage references
<code>eta</code>	<code>VSCConverter.eta</code>	default 0.99	conversion efficiency
<code>loss_percent</code>	<code>VSCConverter.loss_percent</code>	default 0.0	generic loss factor
<code>loss_mw</code>	<code>none</code>	default 0.0	JSON-only fixed loss term
<code>k_vdc</code>	<code>none</code>	default 0.1	JSON-only DC-voltage droop coefficient
<code>pmax_mw, pmin_mw, qmax_mvar, qmin_mvar</code>	same columns	default 0.0	operating capability limits
<code>pRated_mw</code>	<code>VSCConverter.pRated_mw</code>	default 0.0	converter rating
<code>forced_outage_rate, mttr_hr</code>	<code>none</code>	default 0.0	JSON-only reliability metadata
<code>type, vn_ac_kv, vn_dc_kv, k_p, k_q, v_ref_pu, f_ref_hz, controllable, grid-forming / SC fields, t_scheduled_hr</code>	<code>none</code>	not part of current JSON / Excel contract	newer C++ struct fields that are not preserved by either serializer

DCDCConverter : `dc.dcdc_converters[]` / sheet DCDCCConverter

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index, bus_in, bus_out</code>	corresponding columns	required on JSON import; blank Excel <code>index</code> stops table import	stable identifier and input / output DC bus indices
<code>name</code>	<code>DCDCConverter.name</code>	default empty string	converter label
<code>in_service</code>	<code>DCDCConverter.in_service</code>	default <code>true</code>	availability flag
<code>control_mode</code>	<code>DCDCConverter.control_mode</code>	default <code>Voltage</code>	control-mode token <code>Voltage/Power/Droop</code>
<code>p_ref_mw</code>	<code>DCDCConverter.p_ref_mw</code>	default <code>0.0</code>	active-power reference
<code>v_ref_pu</code>	<code>DCDCConverter.v_ref_pu</code>	default <code>1.0</code>	output-voltage reference
<code>sn_mva</code>	<code>DCDCConverter.sn_mva</code>	default <code>0.0</code>	converter rating
<code>vn_in_kv, vn_out_kv</code>	<code>none</code>	default <code>0.0</code>	JSON-only nominal voltage ratings
<code>eta</code>	<code>DCDCConverter.eta</code>	default <code>0.98</code>	conversion efficiency
<code>r_eq_pu</code>	<code>none</code>	default <code>0.0</code>	JSON-only equivalent resistance
<code>pmax_mw, pmin_mw</code>	same columns	default <code>0.0</code>	power limits
<code>k_droop</code>	<code>none</code>	default <code>0.0</code>	JSON-only droop coefficient
<code>mtbf_hours, mtrr_hours</code>	<code>none</code>	default <code>0.0</code>	JSON-only reliability meta-data
<code>controllable, f_switching_hz, t_scheduled_hr</code>	<code>none</code>	not part of current JSON / Excel contract	C++ struct fields that are not preserved by either serializer

DCCircuitBreaker : `dc.dc_circuit_breakers[]` / sheet `DCCircuitBreaker`

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index, bus_from, bus_to</code>	corresponding columns	required on JSON import; blank Excel <code>index</code> stops table import	stable identifier and terminal DC bus indices
<code>name</code>	<code>DCCircuitBreaker.name</code>	default empty string	breaker label
<code>in_service</code>	<code>DCCircuitBreaker.in_service</code>	default <code>true</code>	availability flag
<code>breaker_type</code>	<code>DCCircuitBreaker.breaker_type</code>	default <code>CB</code>	breaker-type token parsed by <code>breaker_type_from_str</code>
<code>closed</code>	<code>DCCircuitBreaker.closed</code>	default <code>true</code>	topology status flag
<code>r_ohm</code>	<code>DCCircuitBreaker.r_ohm</code>	default <code>0.0</code>	DC breaker resistance
<code>rated_voltage_kv, iRated_ka, iBreaking_ka</code>	same columns	default <code>0.0</code>	voltage and current ratings
<code>element_type, element_id</code>	same columns	defaults empty string, <code>0</code>	optional protected-element reference

EnergyRouter : `energy_routers[]` / sheet `EnergyRouter`

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code>	<code>EnergyRouter.index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable router identifier
<code>name</code>	<code>EnergyRouter.name</code>	default empty string	router label
<code>in_service</code>	<code>EnergyRouter.in_service</code>	default <code>true</code>	availability flag
<code>router_type</code>	<code>EnergyRouter.router_type</code>	default empty string	free-form router classification token
<code>num_ports</code>	<code>EnergyRouter.num_ports</code>	default 0	expected number of ports; workbook import may later recompute it from the separate port table
<code>ports[]</code>	sheet <code>EnergyRouterPort</code>	missing JSON array keeps the vector empty	nested port objects are flattened into a dedicated workbook sheet
<code>p_rated_mw</code>	<code>EnergyRouter.p_rated_mw</code>	default 0.0	total router rating
<code>vn_ac_kv, vn_dc_kv</code>	same columns	default 0.0	nominal AC/DC voltage ratings
<code>loss_percent</code>	<code>EnergyRouter.loss_percent</code>	default 0.0	aggregate conversion loss factor
<code>pmax_mw, pmin_mw, qmax_mvar, qmin_mvar</code>	same columns	default 0.0	aggregate operating limits
<code>mtbf_hours, mttr_hours</code>	same columns	default 0.0	reliability metadata preserved in both JSON and workbook
<code>control_mode, power_dispatch_strategy, t_scheduled_hr</code>	<code>none</code>	not part of current JSON / Excel contract	C++ struct fields that are not preserved by either serializer

EnergyRouterPort : nested `energy_routers[].ports[]` / sheet `EnergyRouterPort`

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>router_index</code>	<code>EnergyRouterPort.router_index</code>	workbook-only join key	parent router identifier used because ports are nested in JSON but flattened in Excel
<code>index</code>	<code>EnergyRouterPort.index</code>	required on JSON import; workbook row also needs <code>router_index</code>	stable port identifier within its router
<code>name</code>	<code>EnergyRouterPort.name</code>	default empty string	port label
<code>bus</code>	<code>EnergyRouterPort.bus</code>	required on JSON import	connected AC or DC bus index
<code>port_type</code>	<code>EnergyRouterPort.port_type</code>	default <code>AC</code>	side-type token <code>AC</code> or <code>DC</code>
<code>voltage_level_kv</code>	<code>EnergyRouterPort.voltage_level_kv</code>	default 0.0	nominal port voltage level
<code>p_mw, q_mvar</code>	same columns	default 0.0	present active and reactive transfer assigned to the port
<code>v_pu</code>	<code>EnergyRouterPort.v_pu</code>	default 1.0	local port voltage state or initialization
<code>pmax_mw, pmin_mw, qmax_mvar, qmin_mvar</code>	same columns	default 0.0	operating limits

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>control_mode</code>	<code>EnergyRouterPort.control_mode</code>	default PQ	control-mode token PQ/VF/Droop
<code>p_set_mw, q_set_mvar, v_set_pu</code>	same columns	defaults 0.0, 0.0, 1.0	reference setpoints for the local port controller
<code>in_service</code>	<code>EnergyRouterPort.in_service</code>	default true	availability flag
<code>side, eta</code>	none	not part of current JSON / Excel contract	C++ port fields that are not preserved by either serializer

VirtualPowerPlant : `vpps[]` / sheet VPP

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code>	<code>VPP.index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable VPP identifier
<code>name</code>	<code>VPP.name</code>	default empty string	VPP label
<code>description</code>	none	default empty string	JSON-only free-form description
<code>aggregation_bus</code> / JSON <code>pcc_bus</code>	<code>VPP.pcc_bus</code>	exporter writes <code>pcc_bus</code> ; importer accepts <code>pcc_bus</code> or legacy <code>aggregation_bus</code> ; default 0	reporting / control aggregation bus stored in the C++ <code>aggregation_bus</code> member
<code>in_service</code>	<code>VPP.in_service</code>	default true	availability flag
<code>n_pv_systems, n_wind_turbines, n_battery_systems</code>	same columns	default 0	asset counts preserved in both JSON and workbook
<code>n_ev_chargers, n_controllable_loads</code>	none	default 0	JSON-only aggregation metadata
<code>p_generation_sum_mw, e_storage_sum_mwh, p_load_controllable_mw</code>	none	default 0.0	JSON-only aggregate capability metadata
<code>p_output_mw, q_output_mvar</code>	same columns	default 0.0	present aggregated power output
<code>pmax_mw, pmin_mw</code>	same columns	default 0.0	aggregate operating limits
<code>ramp_up_max_mw_min, ramp_down_max_mw_min</code>	none	default 0.0	JSON-only ramping metadata
<code>mtbf_hours, mttr_hours</code>	none	default 0.0	JSON-only reliability metadata

Microgrid : `microgrids[]` / sheet Microgrid

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index</code>	<code>Microgrid.index</code>	required on JSON import; blank Excel <code>index</code> stops table import	stable microgrid identifier
<code>name</code>	<code>Microgrid.name</code>	default empty string	microgrid label
<code>description</code>	<code>none</code>	default empty string	JSON-only free-form description
<code>in_service</code>	<code>Microgrid.in_service</code>	default <code>true</code>	availability flag
<code>pcc_bus</code>	<code>Microgrid.pcc_bus</code>	exporter writes <code>pcc_bus</code> ; importer accepts <code>pcc_bus</code> or legacy <code>aggregation_bus</code> ; default 0	point-of-common-coupling bus index stored in the C++ <code>aggregation_bus</code> member
<code>internal_buses</code>	<code>none</code>	missing JSON array keeps the vector empty	JSON-only explicit internal bus membership list
<code>operating_mode</code>	<code>Microgrid.operating_mode</code>	default <code>GridConnected</code>	operating-mode token; one of <code>GridConnected</code> , <code>Islanded</code> , or <code>Transition</code>
<code>islanding_capability</code>	<code>Microgrid.islanding_capability</code>	default <code>false</code>	whether the microgrid may island
<code>auto_reconnection</code>	<code>none</code>	default <code>false</code>	JSON-only reconnection automation flag
<code>p_exchange_max_mw</code> , <code>p_exchange_min_mw</code>	Excel keeps only <code>p_exchange_max_mw</code>	default 0.0	aggregate exchange limits with the host grid
<code>p_import_max_mw</code> , <code>p_export_max_mw</code>	<code>none</code>	default 0.0	JSON-only directional exchange limits
<code>p_exchange_mw</code>	<code>Microgrid.p_exchange_mw</code>	default 0.0	present net power exchange at the PCC
<code>total_generation_mw</code> , <code>total_load_mw</code>	same columns	default 0.0	current aggregate generation and demand
<code>total_storage_mwh</code> , <code>capacity_mw</code> , <code>peak_load_mw</code>	<code>none</code>	default 0.0	JSON-only planning meta-data
<code>f_set_hz</code> , <code>v_set_pu</code> , <code>k_droop</code>	<code>none</code>	defaults 50.0, 1.0, 0.0	JSON-only islanded-control targets
<code>area</code>	<code>none</code>	default 0	JSON-only area tag
<code>mtbf_hours</code> , <code>mttr_hours</code>	<code>none</code>	default 0.0	JSON-only reliability meta-data

MobileStorage : `mobile_storage[]` / sheet `MobileStorage`

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>index, bus</code>	corresponding columns	required on JSON import; blank Excel <code>index</code> stops table import	stable identifier and current connected bus
<code>name</code>	<code>MobileStorage.name</code>	default empty string	unit label
<code>in_service</code>	<code>MobileStorage.in_service</code>	default <code>true</code>	availability flag
<code>p_mw</code> , <code>q_mvar</code>	same columns	default 0.0	present dispatch at the current location
<code>p_rated_mw</code>	<code>MobileStorage.p_rated_mw</code>	default 0.0	converter power rating

Field	Excel sheet / column	JSON default / import rule	Meaning / preservation note
<code>pmax_mw, pmin_mw, qmax_mvar, qmin_mvar</code>	none	default 0.0	JSON-only operating limits
<code>e Rated_mwh</code>	<code>MobileStorage.e Rated_mwh</code>	default 0.0	energy capacity
<code>soc_init</code>	<code>MobileStorage.soc_init</code>	default 0.5	initial state of charge
<code>soc_min, soc_max</code>	none	defaults 0.1, 0.9	JSON-only admissible state-of-charge bounds
<code>eta_charge, eta_discharge</code>	none	default 0.95	JSON-only efficiency metadata
<code>max_cycles, current_cycles, soh</code>	none	defaults 5000, 0, 1.0	JSON-only lifecycle / health metadata
<code>is_mobile</code>	none	default true	JSON-only flag distinguishing mobile vs. stationary storage representation
<code>status</code>	<code>MobileStorage.status</code>	default <code>Stationary</code>	mobility-state token; one of <code>Stationary</code> , <code>InTransit</code> , or <code>Deployed</code>
<code>current_location</code>	none	default empty string	JSON-only free-form location string
<code>target_bus</code>	none	default 0	JSON-only relocation target
<code>e_consumption_mwh_kwh, max_travel_distance_kwh</code>	none	default 0.0	JSON-only transportation-energy metadata
<code>mtbf_hours, mttr_hours</code>	none	default 0.0	JSON-only reliability metadata

10.5 Cross-Reference

The only remaining detailed dictionaries not repeated in this section are the ones already written in Section 02a_io_contracts:

1. `AsynchronousMotor`
2. the optional `three_phase_ac` metadata container
3. `ThreePhaseBus`, `ThreePhaseLine`, `ThreePhaseTransformer`, `ThreePhaseLoad`, `ThreePhaseGenerator`, and `ThreePhaseExternalGrid`
4. result-sheet enum and flag conventions

11 Output and Report Field Dictionaries

This section applies the same field-level dictionary template to the current export artifacts. Section 02a_io_contracts already catalogs which workbooks and JSON serializers exist; the tables below make the output-side contract explicit at workbook-column and JSON-key level.

11.1 Export Families

Artifact	Export path	Direction	Current note
<code>PowerFlowResult</code> dedicated workbook	source-only workbook helper	export only	writes the <code>Results_*</code> sheets only when optional Excel support is compiled.
<code>PowerFlowResult</code> compact workbook	<code>save_results_compact.xlsx</code>	export only	rewrites input sheets and appends <code>result_*</code> columns to selected tables, plus <code>Results_Summary</code> .

Artifact	Export path	Direction	Current note
PowerFlowResult JSON	power_flow_result_to_json / save_power_flow_result_json	export only	serializes convergence summary plus AC/DC bus states, AC/DC branch flows, VSC transfers, and DC/DC transfers.
ACOPFResult dedicated workbook	save_opf_results_xlsx	export only	writes OPF_* sheets only.
ACOPFResult compact workbook	save_opf_results_compact_xlsx	export only	rewrites input sheets but still emits separate OPF_* sheets; no opf_* columns are appended into input tables.
ACOPFResult JSON	opf_result_to_json / opf_result_from_json	bidirectional	serializes scalar summary, solver_path, bus / generator / converter vectors, enhanced dispatch objects with maps, legacy dispatch arrays, and OPF profiling block.
CarbonAnalysisResult workbook	save_carbon_results_xlsx	export only	writes AC/DC load and bus sheets, AC/DC branch-loss sheets, converter-loss sheets, storage / EnergyRouter sheets, and block-style Summary.
CarbonAnalysisResult JSON	carbon_result_to_json / carbon_result_from_json	bidirectional	preserves the richer tracing maps and the hybrid AC/DC carrier results that the workbook flattens.
Short-circuit and ONR artifacts	available via module headers; not forwarded from hacdcpf.hpp	hacdcpf/analysis/short_circuit.hpp, hacdcpf/network_reconfiguration/topology_analysis.hpp	Fully implemented and tested. No dedicated workbook serializer exists; results are consumed in-memory or logged. Output field contracts are documented in those module headers.

11.2 Result Semantic Conventions

Semantic aspect	Current rule	Main artifacts	Engineering note
Electrical units	Active power uses MW, reactive power uses Mvar, AC / DC voltage magnitudes use pu, AC voltage angle uses degree, and current stays in pu unless the field name explicitly says ka .	PF / OPF workbooks and JSON	Workbook and JSON field names usually encode the unit directly, for example pg_mw , va_deg , loss_mw , and matrix_residual .
Carbon units	Carbon intensity uses tCO2/MWh ; total emissions use tCO2 .	Carbon workbook and JSON	The JSON / workbook field names use the suffixes _tco2_mwh and _tco2 ; no implicit mass-unit conversion is applied by the exporter.
Vector order	Numerical result vectors follow solver-side component order, not sparse physical ids. vm , va , dpd_mw , and dqd_mvar follow AC-bus order; vdc follows DC-bus order; pg_mw , qg_mvar follow AC-generator order; pac_mw , qac_mvar follow VSC order.	PF / OPF JSON and workbook exports	Consumers should treat workbook rows and JSON vectors as aligned to the source component arrays unless a sheet explicitly says otherwise.
Row identity	Workbook result sheets copy source model index values when a source component exists; Results_BranchFlow.index is currently ACBranch::index .	PF workbooks and PF JSON	For branch-flow joins, use the source component index plus terminal buses; do not assume row number equals component id.
Fallback values	When a solver vector entry is absent or out of range, exporters write deterministic defaults: typically 0.0 for powers, angles, counters, and residual-like scalars, or 1.0 for voltage magnitudes.	PF / OPF workbooks	This keeps workbook shapes stable, but it means a consumer must distinguish “solver wrote a fallback” from “the physical result is exactly zero” using convergence and artifact context.

Semantic aspect	Current rule	Main artifacts	Engineering note
JSON missing keys	JSON importers keep absent arrays or object maps empty and default absent scalar keys to false, zero, or empty string according to the target field type.	OPF / carbon JSON	The JSON contracts are intentionally sparse-friendly; round-trip consumers should not require every optional key to be present.
Echoed vs solved values	Some exported fields are metadata echoes or derived annotations rather than true solver states: storage <code>soc</code> comes from input <code>soc_init</code> ; router port sums come from stored port structs; <code>Results_DC Branch</code> is recomputed from solved voltages and branch resistance.	PF workbooks	Consumers should not infer that every result-sheet field belongs to the nonlinear state vector.
Boolean spelling	PF / OPF workbook booleans use the standard workbook token form produced by <code>bool_str</code> ; carbon <code>Summary</code> uses lowercase <code>true/false</code> .	Result workbooks	Downstream parsers should normalize workbook booleans instead of assuming one literal spelling.
Conditional sheets	Some result sheets are pre-created even when no rows are written. <code>OPF_Converter</code> is the main example.	PF / OPF workbooks	Empty sheets are part of the current export contract and do not by themselves indicate an error.
Object-map semantics	Carbon contribution maps are exported as JSON object maps keyed by internal carbon-source id rather than positional arrays.	Carbon JSON	Map-key order is not semantically meaningful; workbook export intentionally flattens these maps away.